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Flood

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(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **METHOD FOR DEPLOYING LARGE NUMBERS OF WASTE CONTAINERS IN A WASTE COLLECTION SYSTEM**

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(72) Inventor: **Christopher Flood, Carol Stream, IL (US)**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

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US 2024/0211849 A1 Jun. 27, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/473,166, filed on Sep. 13, 2021, now Pat. No. 11,640,575, which is a (Continued)

(51) **Int. Cl.**

G08B 21/00 (2006.01)
G06K 7/00 (2006.01)
G06K 7/10 (2006.01)
G06K 19/14 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **G06Q 10/0833** (2013.01); **G06K 7/0004** (2013.01); **G06K 7/10297** (2013.01); (Continued)

(58) **Field of Classification Search**

CPC .. **G06Q 10/0833**; **G06Q 10/08**; **G06K 7/0004**; **G06K 7/10297**; **G06K 19/145**; **G06V 10/10**

(Continued)

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Primary Examiner — Mark S Rushing

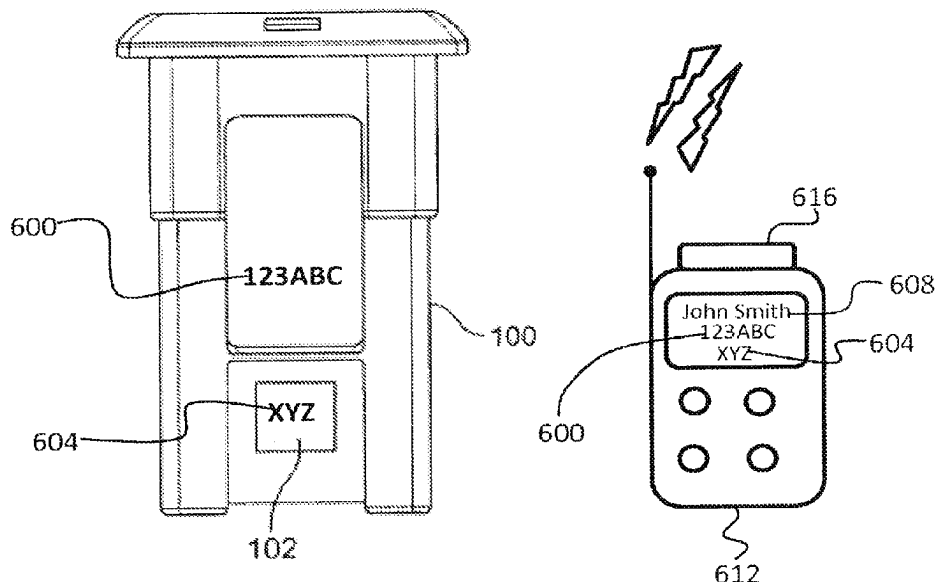
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(57)

ABSTRACT

A method of redeploying waste containers in a waste collection system with an active or passive signal means requires a user to aim an optical character recognition reader at a waste container comprising a unique optical waste container identifier. The optical character recognition reader reads unique waste container optical identifier. The unique waste container optical identifier is converted to a unique electronically formatted waste container identifier. A unique radio frequency signal identifier is stored on a radio frequency emitting tag and associated with the unique electronically formatted waste container identifier. The radio frequency emitting tag is attached to the waste container.

7 Claims, 20 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/829,449, filed on Mar. 25, 2020, now Pat. No. 11,144,736, which is a continuation of application No. 15/011,940, filed on Feb. 1, 2016, now Pat. No. 10,635,864, which is a continuation of application No. 13/895,138, filed on May 15, 2013, now Pat. No. 9,251,388.

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G06Q 10/08 (2023.01)

G06Q 10/0833 (2023.01)

G06V 10/10 (2022.01)

(52) **U.S. Cl.**

CPC **G06K 19/145** (2013.01); **G06Q 10/08** (2013.01); **G06V 10/10** (2022.01)

(58) **Field of Classification Search**

USPC 340/5.81

See application file for complete search history.

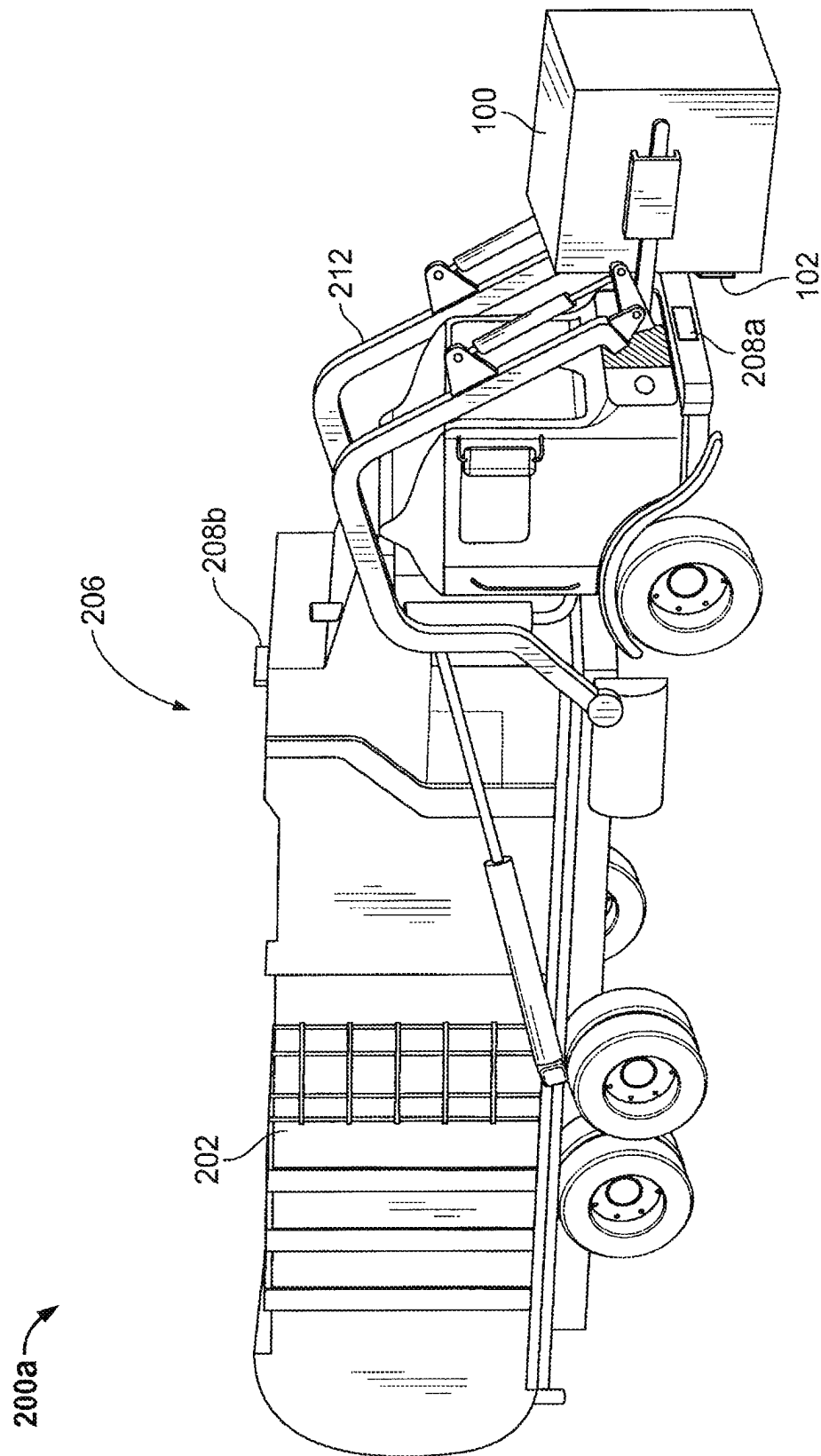
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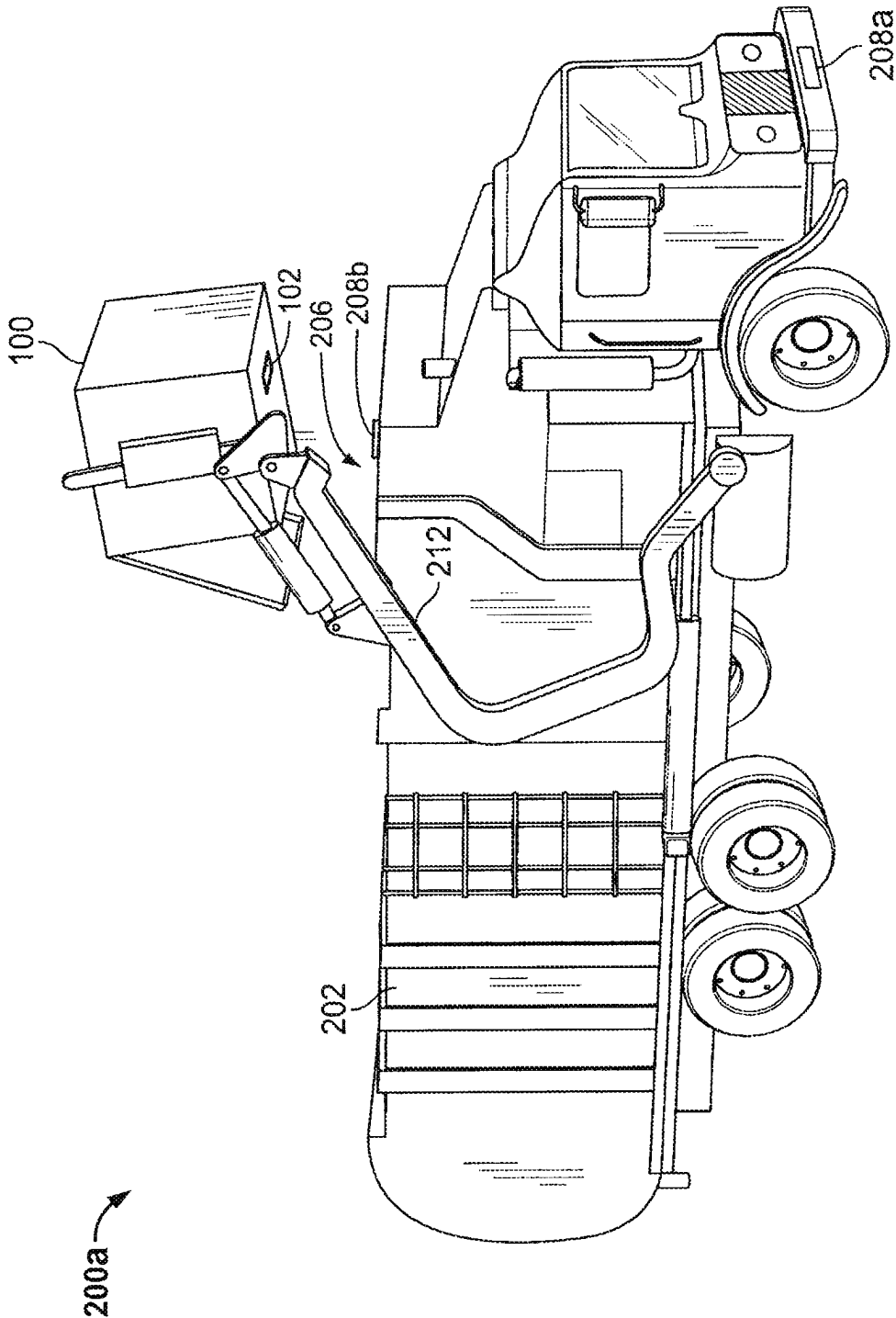


FIG. 2

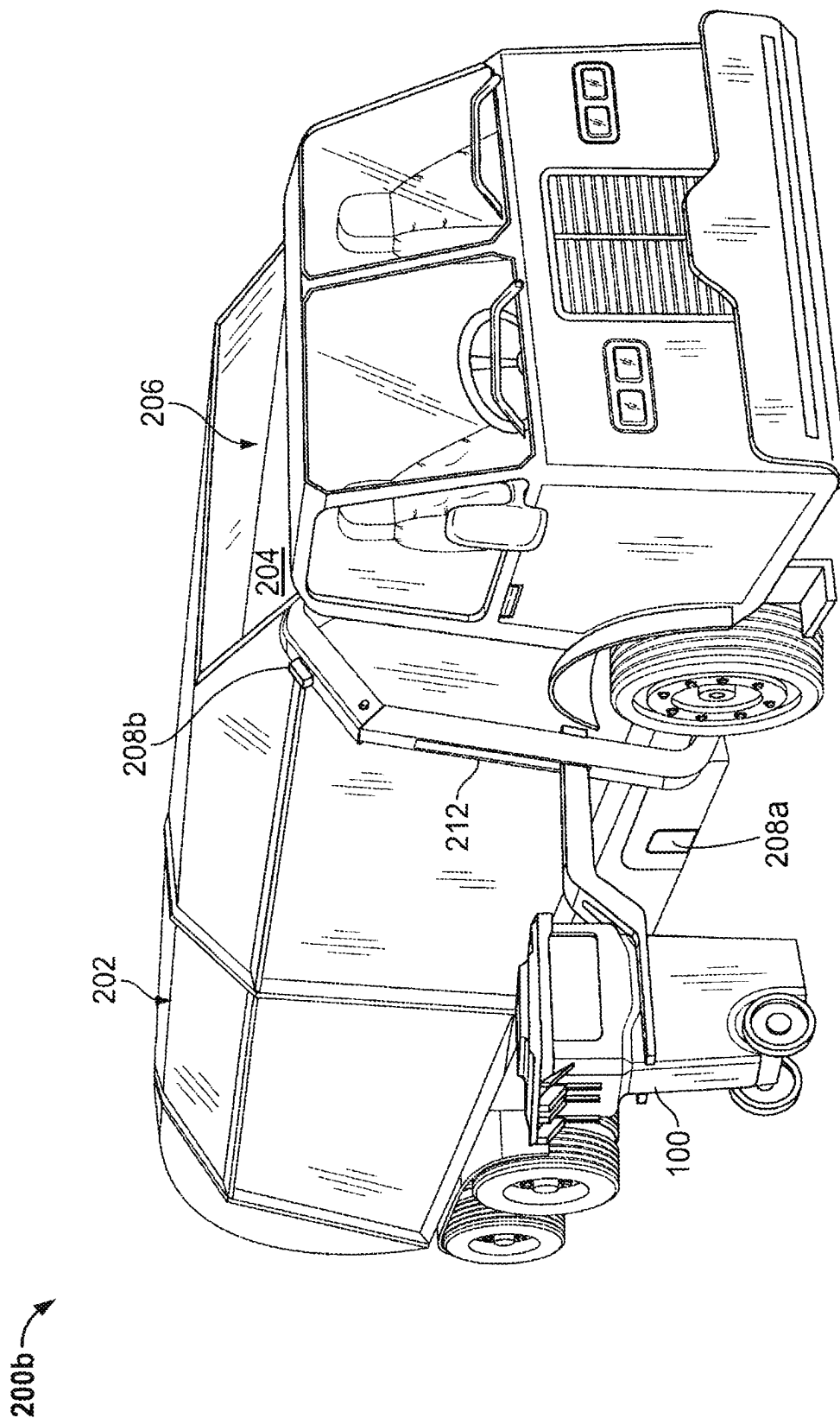


FIG. 3

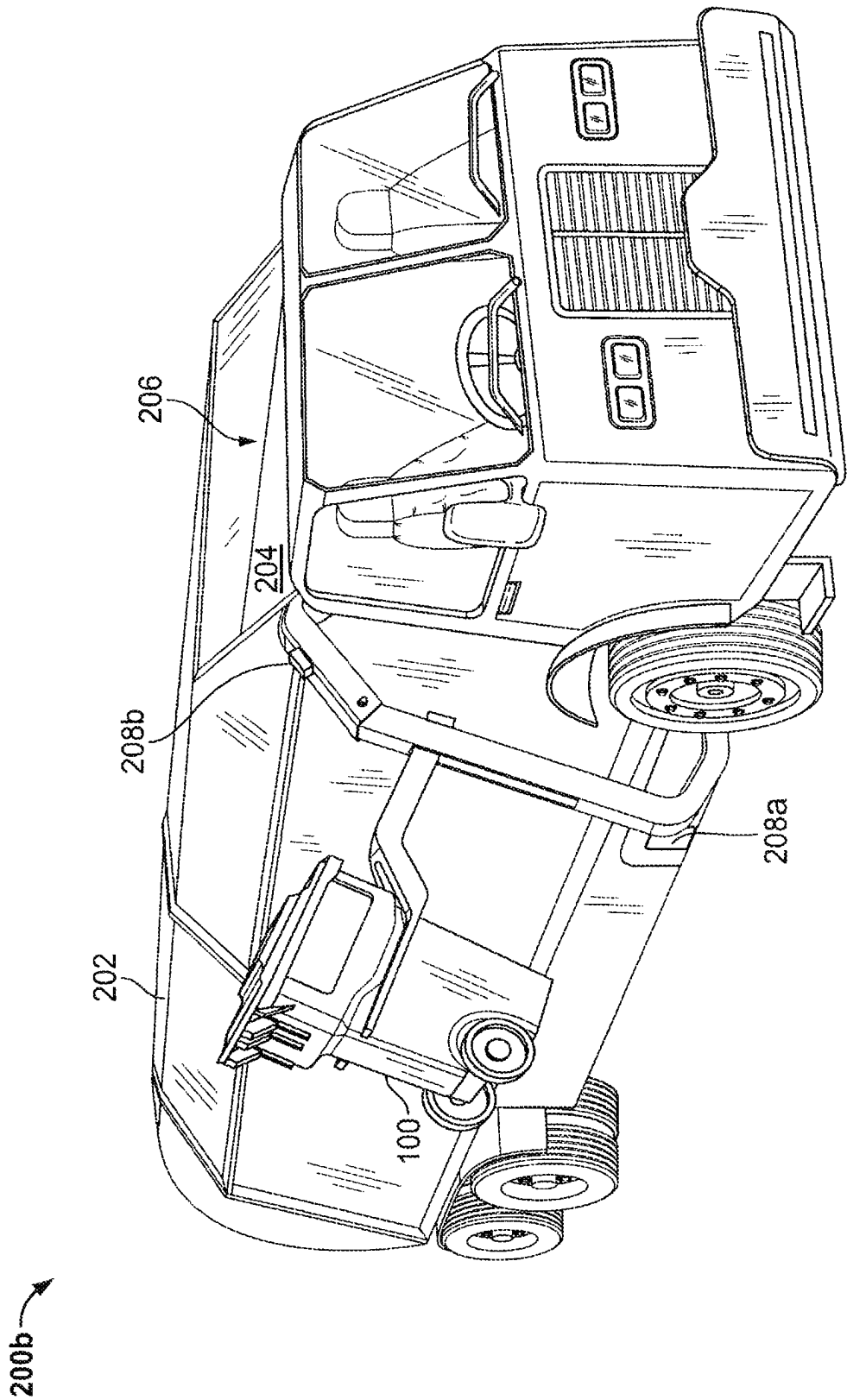


FIG. 4

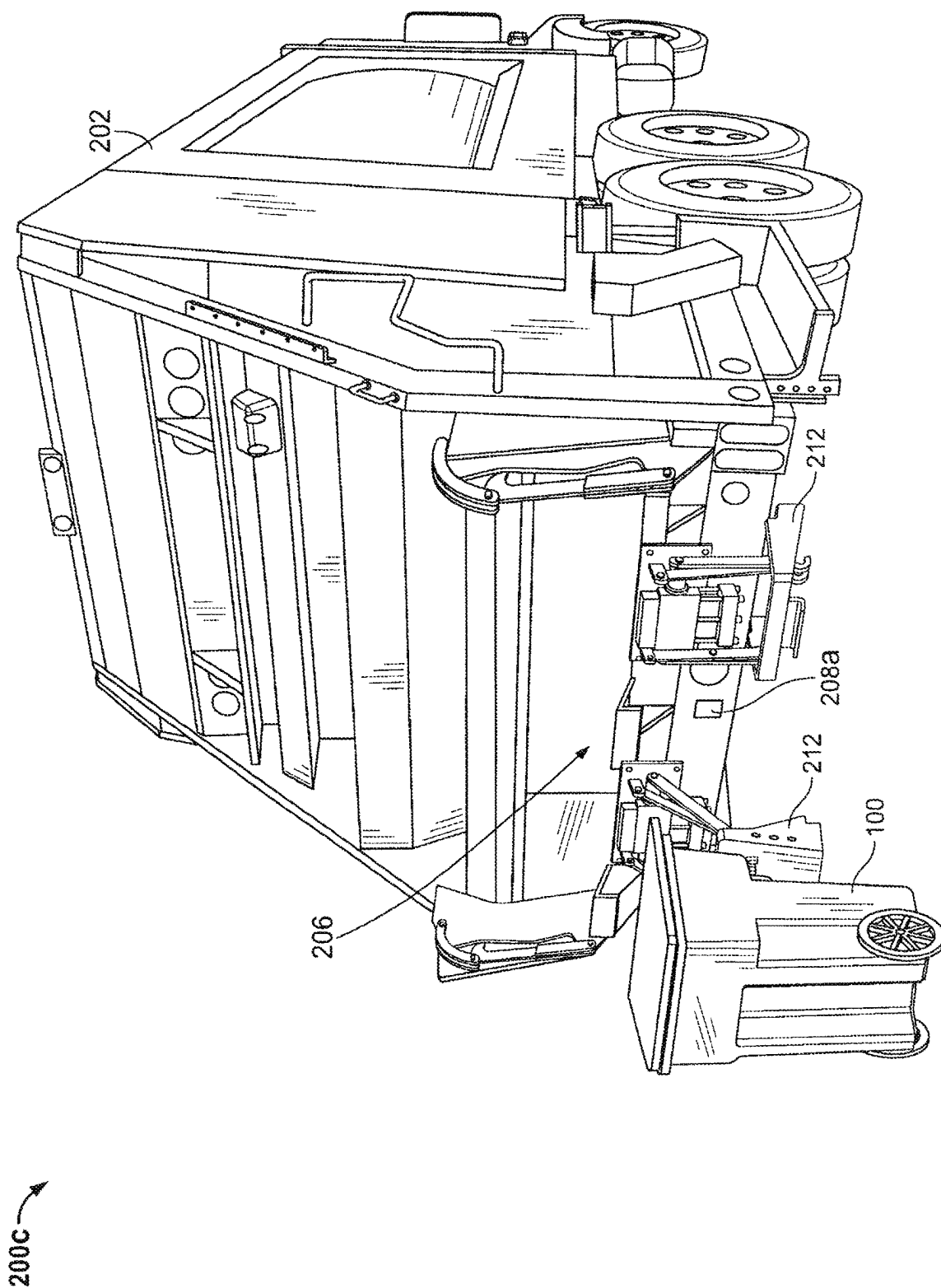


FIG. 5

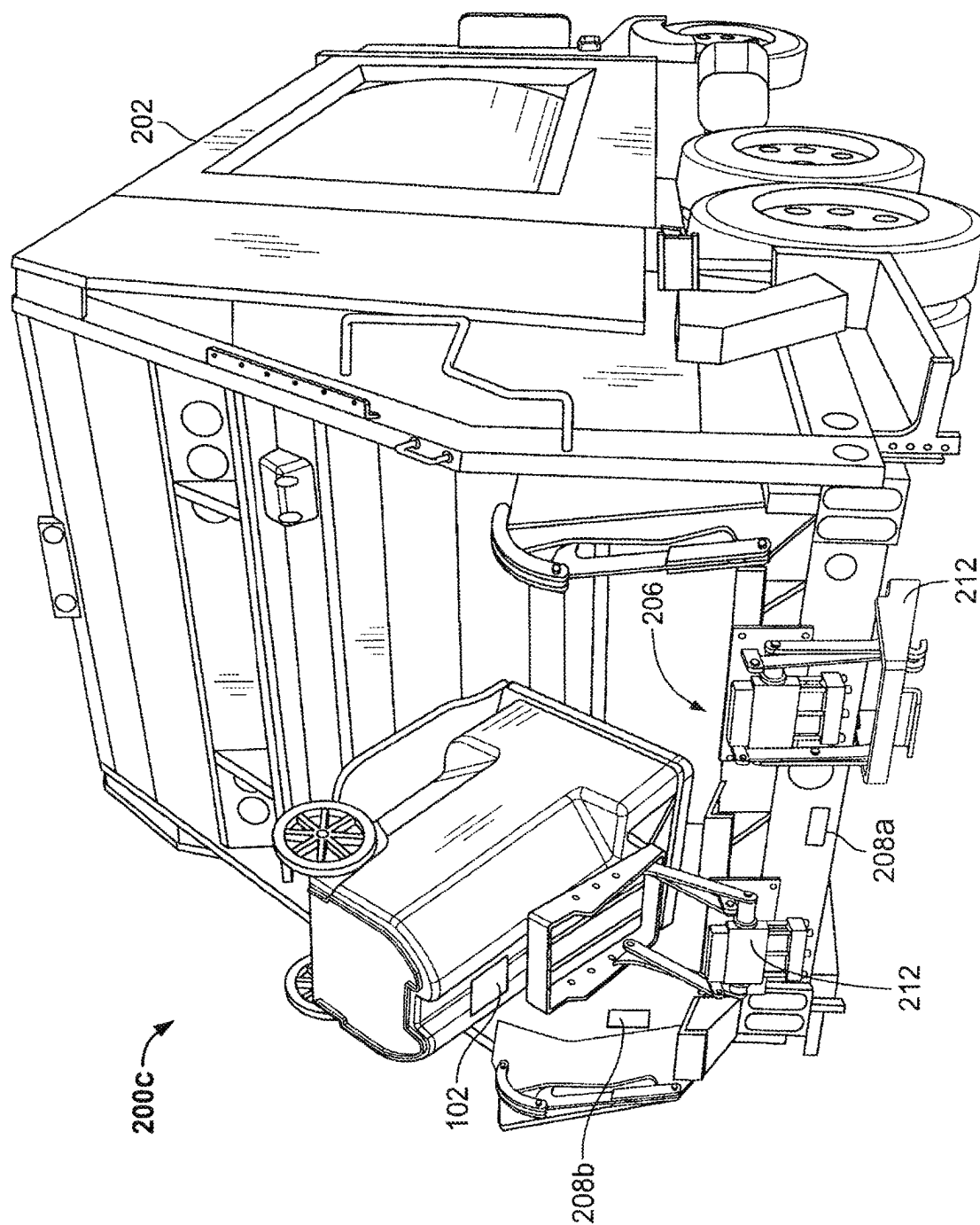


FIG. 6

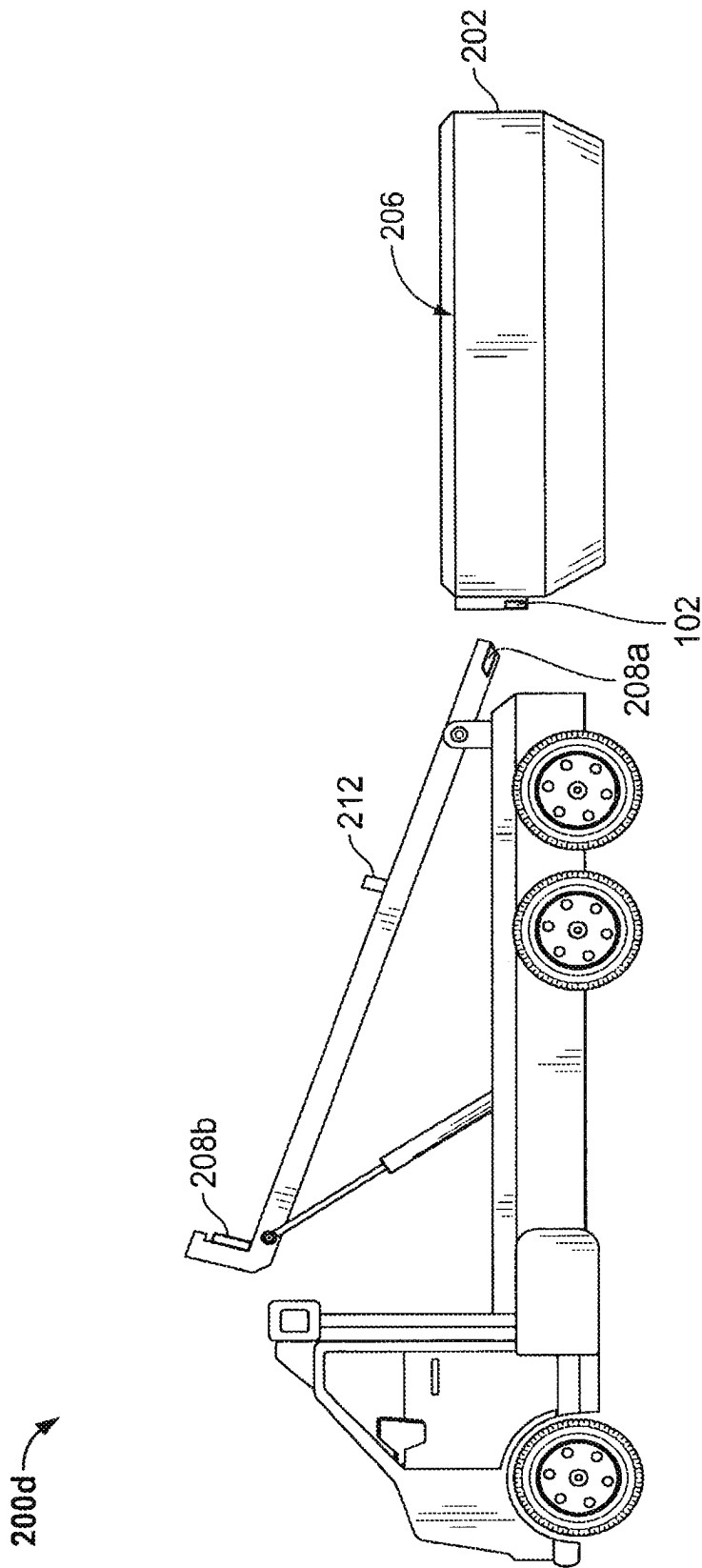
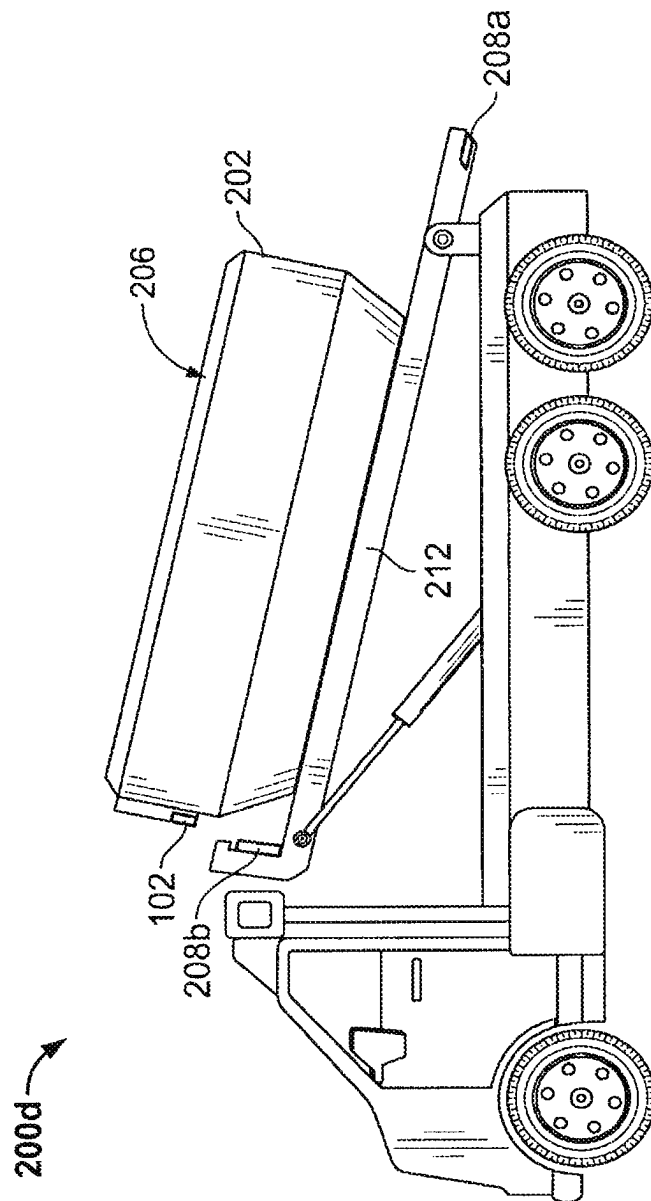


FIG. 7



 NATIONAL SCIENCE FOUNDATION

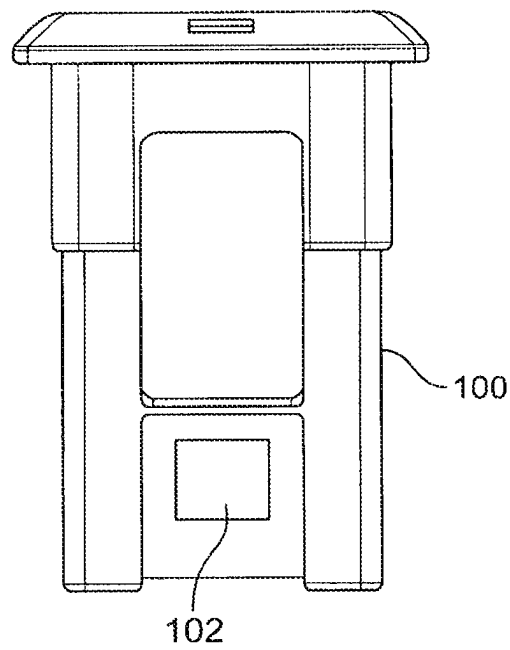
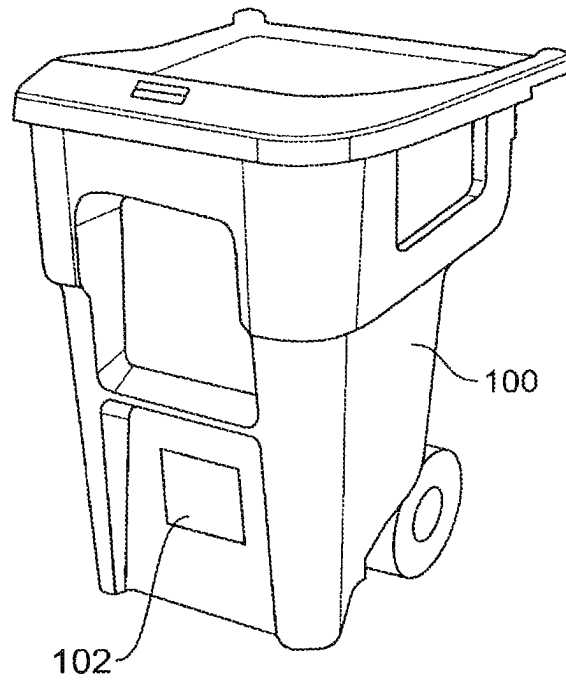


FIG. 9

FIG. 10

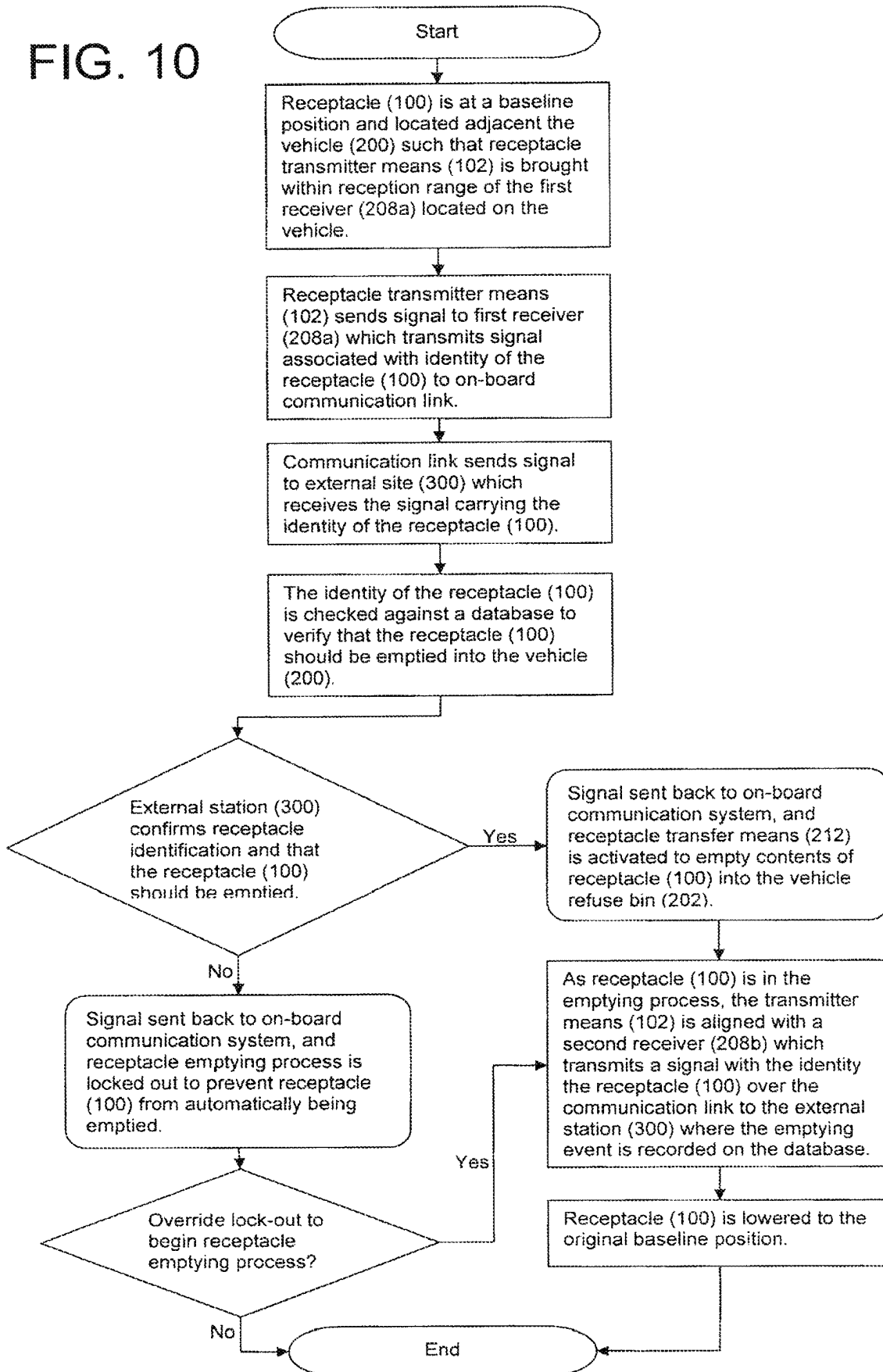


FIG. 11

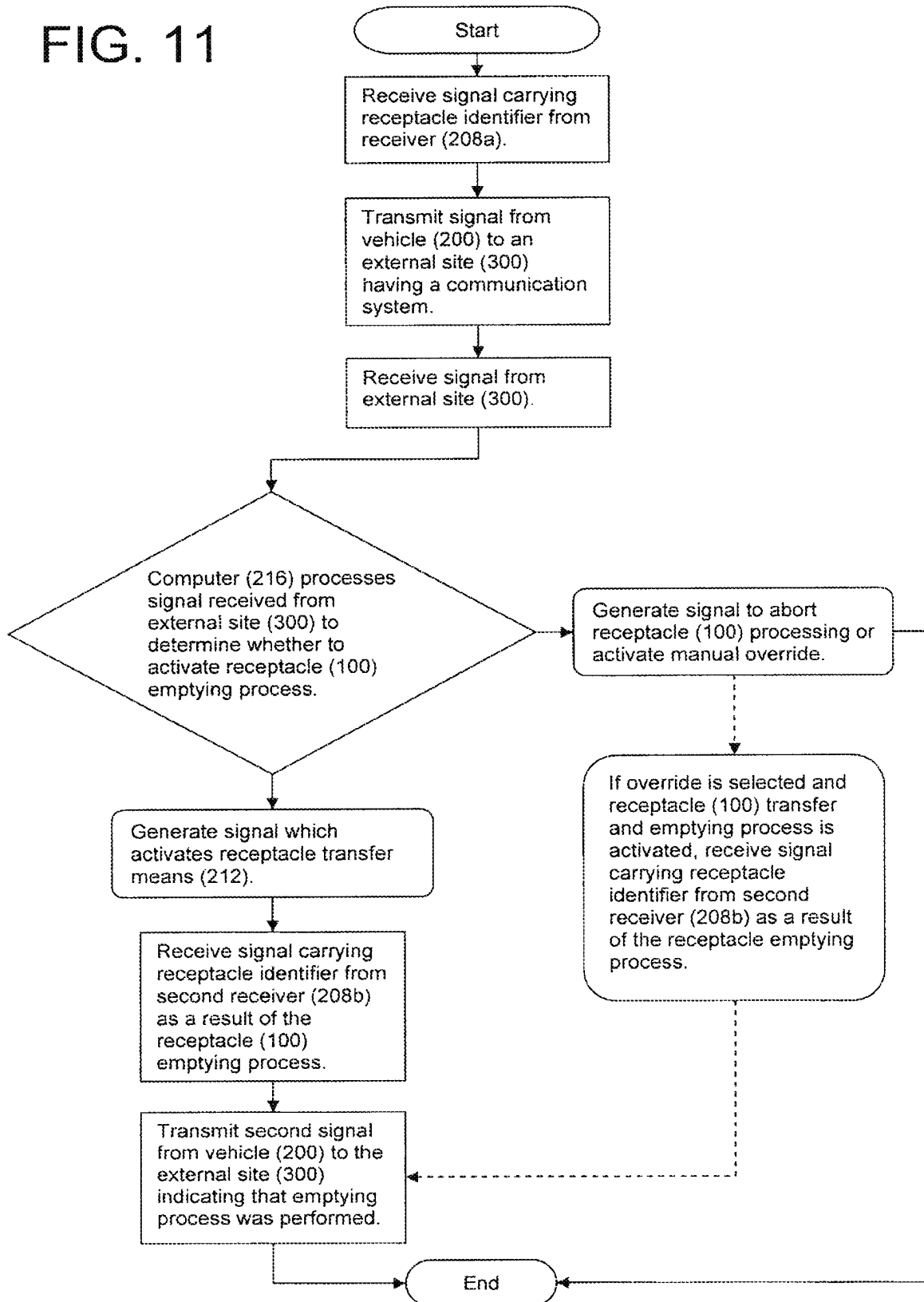


FIG. 12

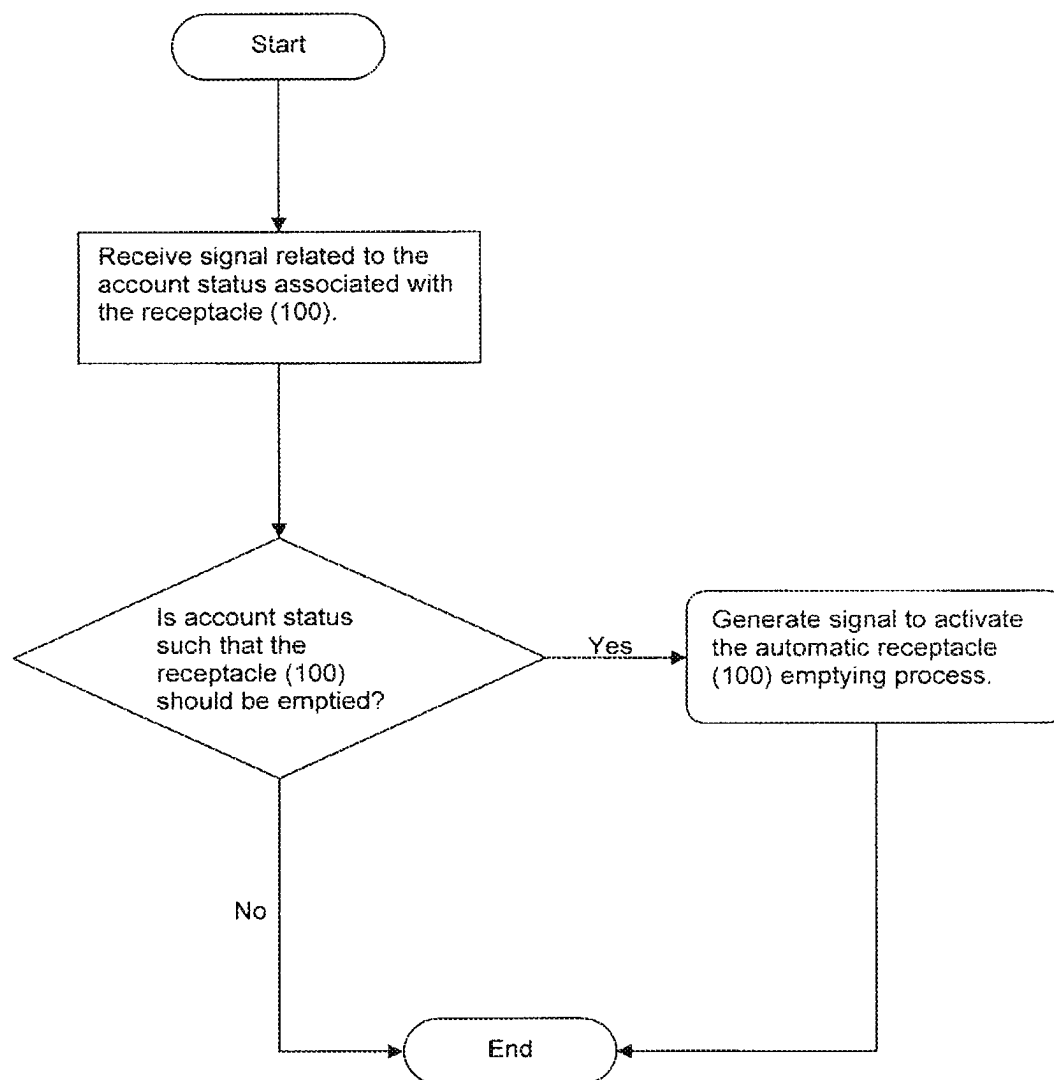


FIG. 13

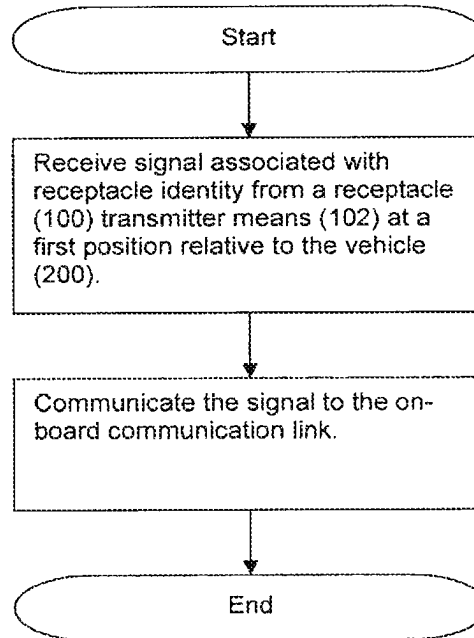


FIG. 14

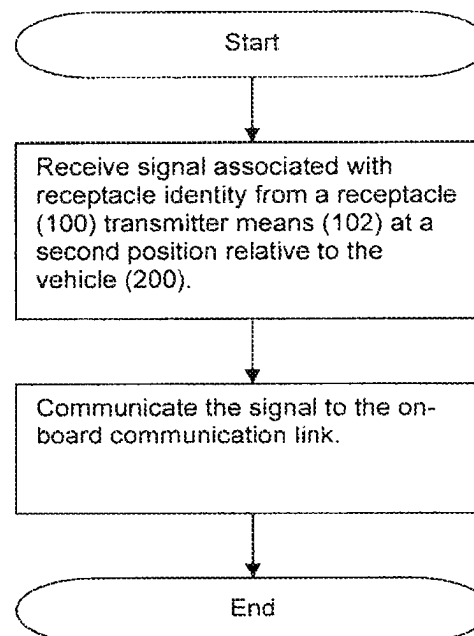


FIG. 15

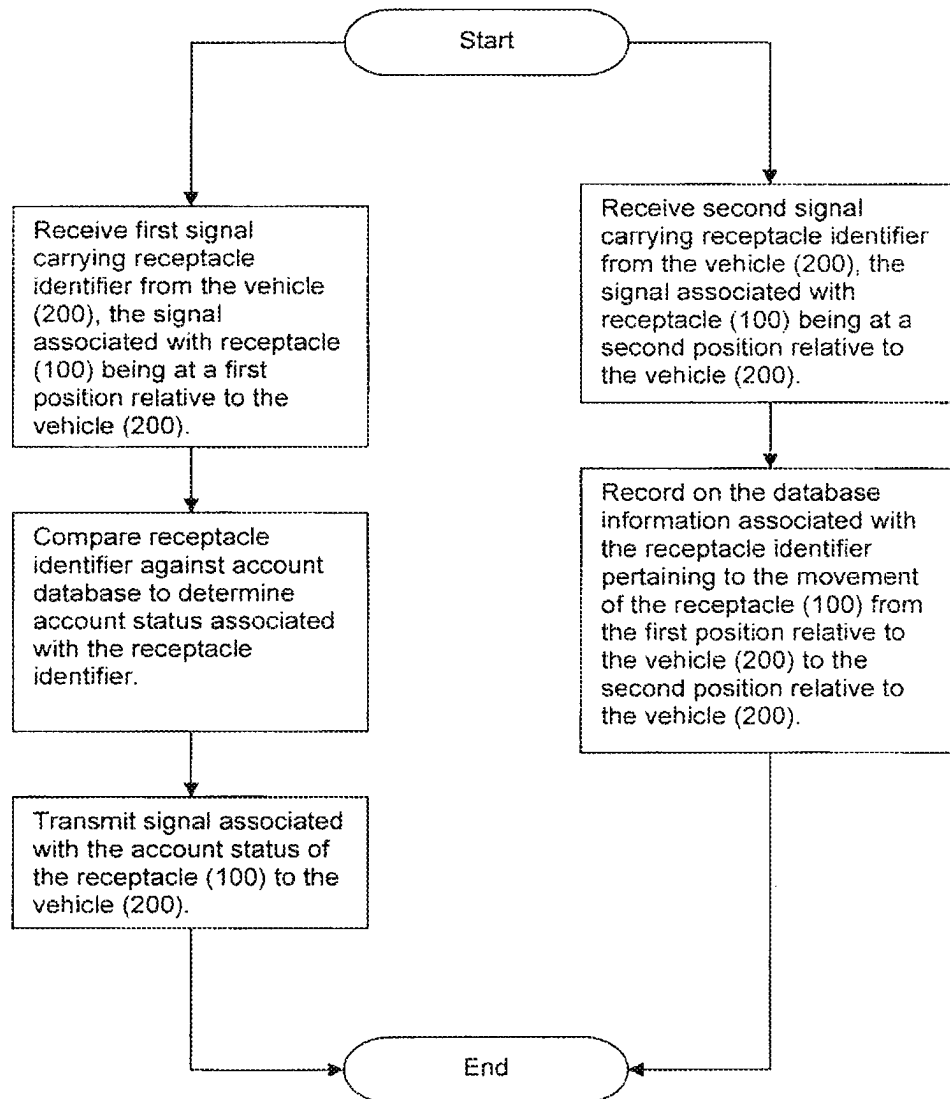


FIG. 16

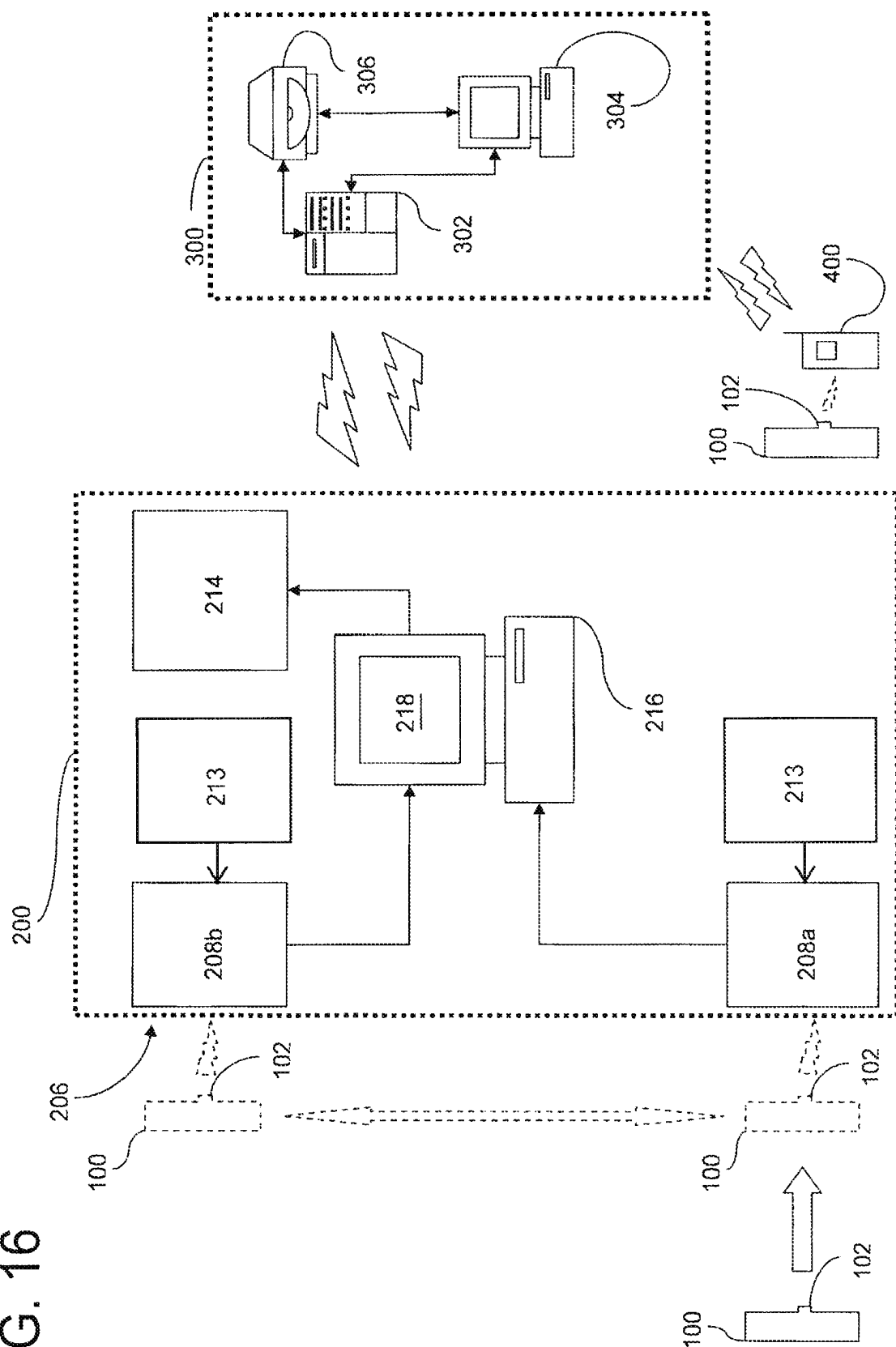
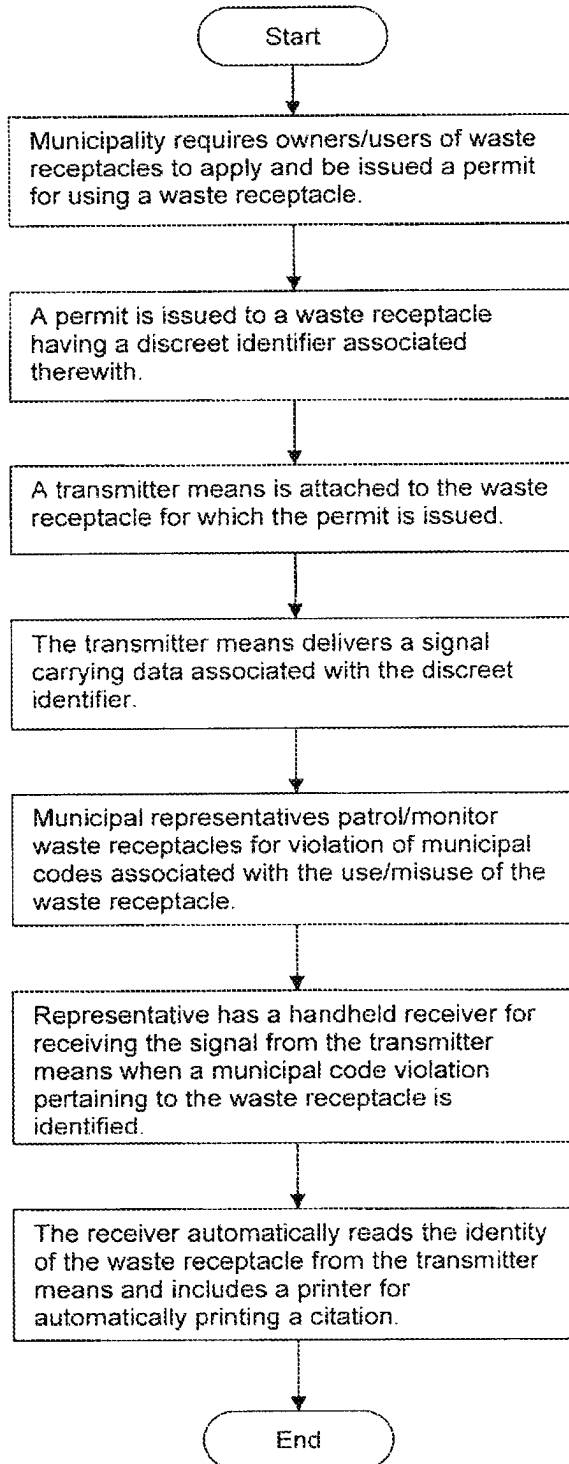


FIG. 17



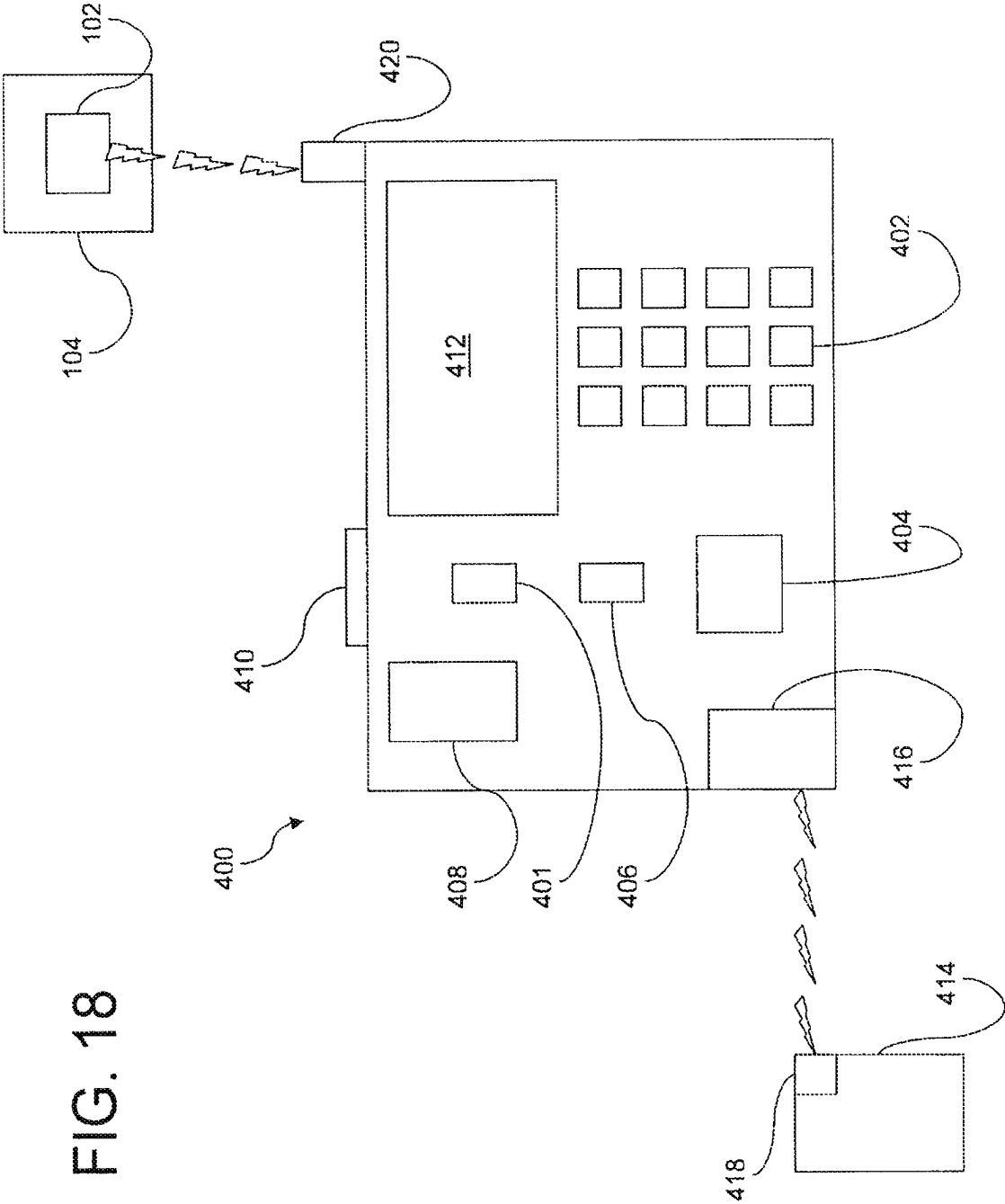
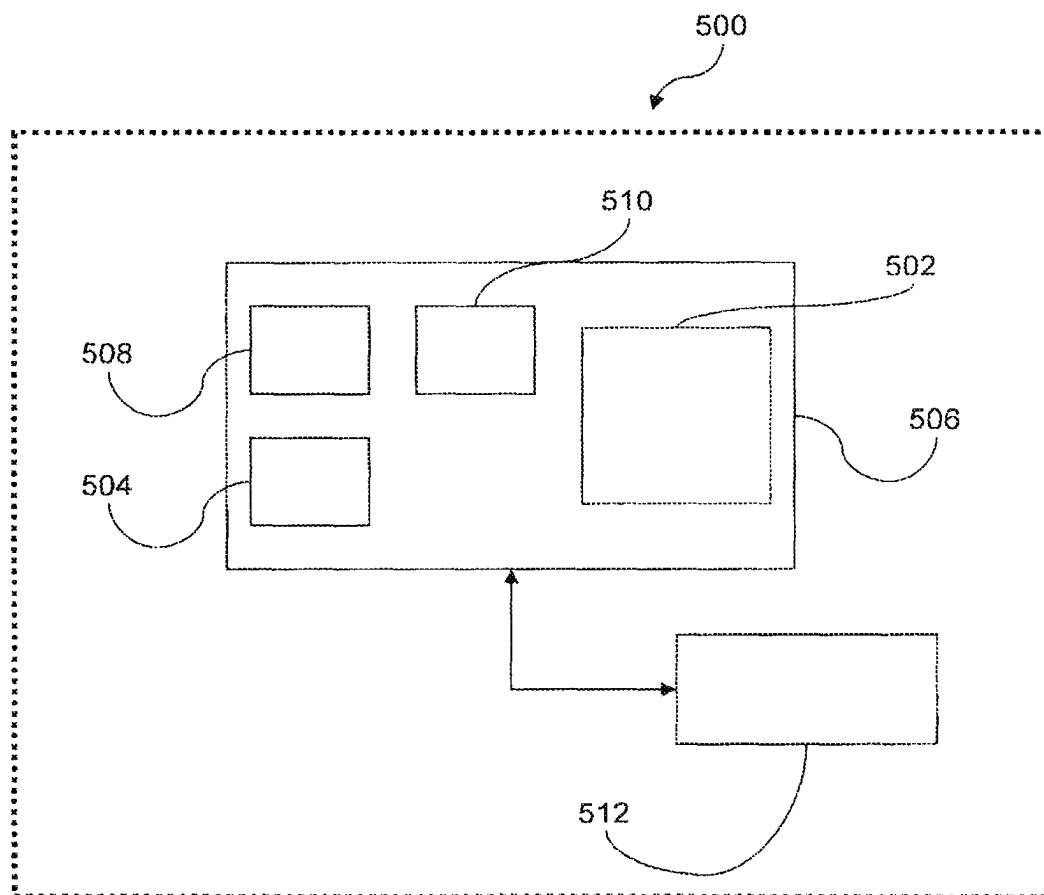


FIG. 19



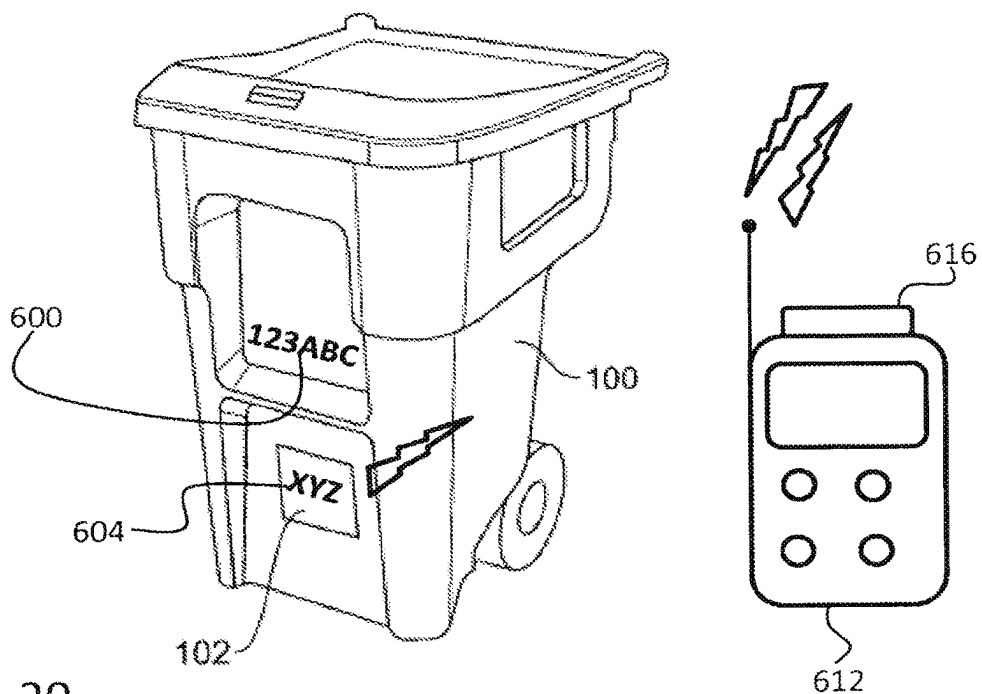


FIG. 20

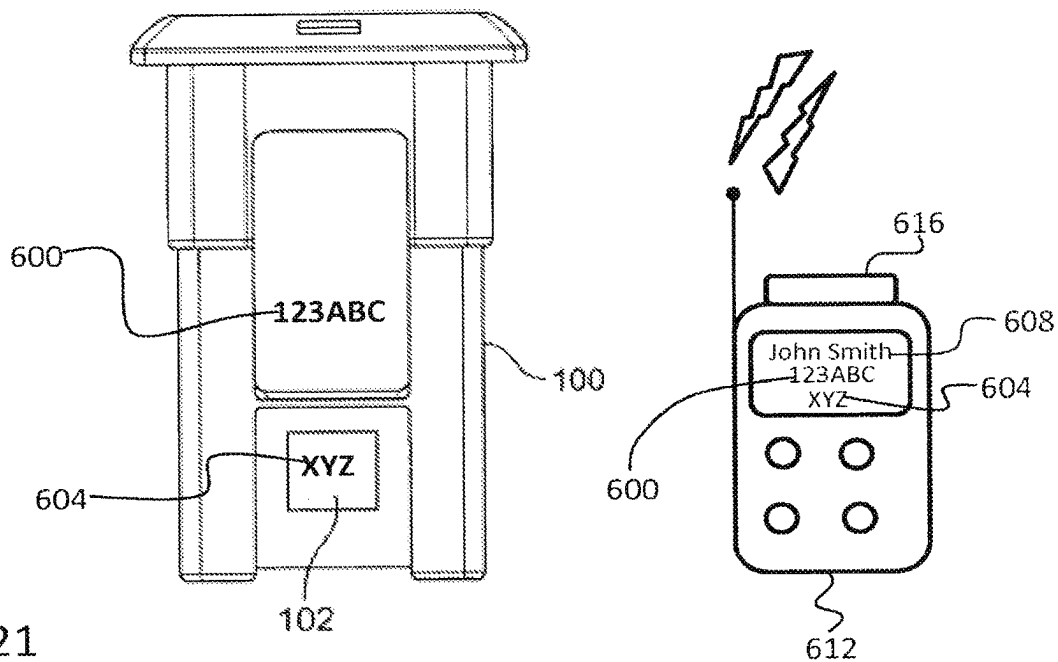


FIG. 21

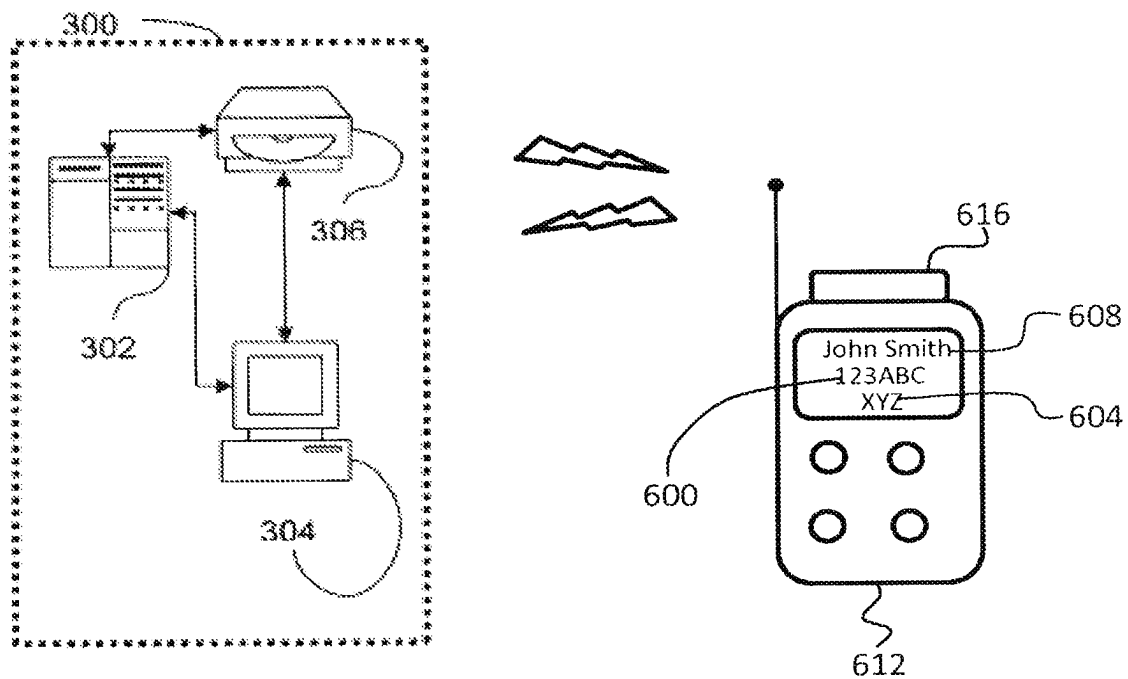


FIG. 22

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METHOD FOR DEPLOYING LARGE NUMBERS OF WASTE CONTAINERS IN A WASTE COLLECTION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of application Ser. No. 13/895,138, now U.S. Pat. No. 9,251,388.

FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

N/A

TECHNICAL FIELD

The invention relates to refuse and waste removing and hauling. More particularly, the present invention relates to a method and apparatus for deploying and/or redeploying waste containers in a waste removing and hauling system.

BACKGROUND OF THE INVENTION

The collection and transportation of trash and recyclables from residential, commercial, industrial and large residential facilities is a major industry in the United States and throughout the civilized world. Typically, trash and recyclables are accumulated and temporarily stored in waste material receptacles such as trash cans and dumpsters. When filled, or at regularly scheduled intervals, trash and recyclables from the containers are transported for the eventual recycling, incineration and/or disposal into landfills.

Customers typically pay for trash and recyclables removal services based on the amount of trash and recyclables removed and the number of trash and recyclables pickups over a period of time. The compacting of trash and recyclables at a customer's location typically reduces the number of pick ups. A successful trash and recyclables compactor is disclosed in US 6A 12A06, titled Trash Compactor, and Owned by Advanced Custom Engineered Systems & Equipment, Inc., Hanover Park, Illinois.

These industrial, commercial and large residential bins and compactors are collected from different locations and hauled to a central location. Normally, those hauling the trash and recyclables are sent from a central location and dispatched to the different locations. In practice, paper logs or schedules document the hauler's runs (e.g., trash and recyclables to pick-up, trash and recyclables being picked-up, and trash and recyclables picked-up). The haulers are given their routes in person or over the phone. The haulers, in turn, keep in touch with the central location generally by cell phone or radio.

For large organizations this can be a very complicated task as there are many haulers and many customers needing their trash and recyclables collected, picked-up and hauled away. In addition, commercial, industrial and large residential (e.g., condos and apartment buildings) trash and recyclables compactors and balers must be monitored for maintenance and repair. This too requires time and energy for the haulers and/or representatives (of the service provider) to monitor and inspect.

It should also be recognized that these industrial, commercial and large residential bins, balers and compactors require both period maintenance and emergency demand repair services. Normally, those repairing the equipment are sent from a central location and dispatched to the different

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locations. In practice, paper logs or work orders document the repairperson's time (e.g., drive time, time spent performing the repairs, parts and materials used, etc.). The repair companies use a variety of management tools. For example, some are given their routes in person or over the phone. The service providers, in turn, keep in touch with the central location generally by cell phone or radio.

For large organizations this can be a very complicated to coordinate and to verify that the charges for these services are fair and accurate as there are many service providers and many customers needing their compactors, bins and balers repaired. In addition, commercial, industrial and large residential (e.g., condos and apartment buildings).

In addition, it must be recognized that trash and recyclables compactors, balers and bins must further be monitored for maintenance and repair.

Methods of improving the refuse collection are disclosed in commonly assigned U.S. Patent Application Publication No. 2008-0197194 A1, published on Aug. 21, 2008; U.S. Patent Application Publication No. 2008-0198021 A1, published on Aug. 21, 2008; and U.S. Patent Application Publication No. 2008-0202357 A1, published on Aug. 28, 2008. These publications are hereby incorporated by reference as if fully set forth herein, and generally disclose systems for communicating with receptacles, etc.

One on-going advancement in the field of refuse removal is directed to improving communication between the vehicles making refuse pick-ups (emptying receptacles) and the central station or dispatcher. Currently, the dispatcher may be in contact with the vehicle via radio or telephone transmission. However, it is not cost-effective or feasible for the dispatcher to be in constant contact with every vehicle out in the field. Thus, it is impossible for the dispatcher to relay account information associated with each receptacle to a vehicle. It would be advantageous to provide such information to the vehicle to prevent pick-up and emptying of receptacles owned or managed by entities delinquent in their payment of invoices. Also, a given account may have special instructions, such as an additional oversized pick-up, for a discreet single day or event. The dispatcher currently has no way of assuring that the special instructions are provided to the vehicle in a timely manner.

Furthermore, there is on-going and growing concern in major municipalities with controlling several aspects refuse collection. For instance, citizens or users will often engage in activity with respect to refuse receptacles that violates municipal codes. Some of this is caused by simply overloading receptacles. Other times, a user may fall behind in payment of bills for removing refuse and waste, and the receptacles and surrounding areas will become over loaded with refuse. This provides a haven for vermin such as rats. It may also cause damage to the lane on which the receptacles are located.

Absent constant patrol of back lanes, municipalities often have no idea that these conditions exist until it is too late. Even when patrolling the back lanes, it is difficult to impossible to determine which receptacles belong to which owners/addresses. Finally, when a positive identification of the offending receptacle is able to be determined the process consumes so much time that the municipal employee is only able to investigate but a handful of the many violations that occur at any one time. Thus, city officials need a method that will facilitate receptacle identification while at the same time providing a simplified method of issuing citations to the proper entity responsible for the code violation.

The problems discussed above are being addressed by the industry. However, another problem remains and that is to

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deploy or redeploy large numbers of waste containers among the hundreds or thousands of users/citizens participating in waste refuse collection system. Currently, there is no automated method of outfitting waste containers with suitable tracking mechanisms, normally an active or passive radio frequency emitting identification tag. Presently, there are literally millions of waste containers in use throughout the country that are not outfitted with such radio frequency tags and no accurate, cost-effective method or system to deploy, outfit, to retrofit the waste containers with an active or passive radio frequency identification tag.

The present invention is provided to solve the problems discussed above and other problems, and to provide advantages and aspects not provided by prior waste and refuse collection systems of this type. A full discussion of the features and advantages of the present invention is deferred to the following detailed description, which proceeds with reference to the accompanying drawings.

The present invention is provided to solve the problems discussed above and other problems, and to provide advantages and aspects not provided by prior deployment systems of this type. A full discussion of the features and advantages of the present invention is deferred to the following detailed description, which proceeds with reference to the accompanying drawings.

SUMMARY OF THE INVENTION

One aspect of the present invention is directed to a method of redeploying waste containers in a waste collection system with an active or passive signal means. The method comprises the following steps: (1) aiming an optical character recognition reader at a waste container comprising a unique optical waste container identifier; (3) reading the unique waste container optical identifier using the optical character recognition reader; (4) converting the unique waste container optical identifier to a unique electronically formatted waste container identifier; (5) associating a unique radio frequency signal identifier stored on a radio frequency emitting tag with the unique electronically formatted waste container identifier; and (6) attaching the radio frequency emitting tag to the waste container.

The first aspect of the present invention may include one or more of the following features and or steps, alone or in any reasonable combination. The method may further comprise the step of storing the unique electronically formatted waste container identifier on the optical character recognition reader. The method may further comprise the step of reading the unique radio frequency signal identifier from the radio frequency emitting tag. The method may further comprise the step of activating a software module for associating the unique radio frequency signal identifier with the unique electronically formatted waste container identifier stored on the optical character recognition reader. The method may further comprise the step of activating a software module for associating the unique radio frequency signal identifier with personal user data of a user of the waste container. The method may further comprise the step of activating a software module for generating a unique database entry comprising the unique radio frequency signal identifier and the personal user data associated therewith. The optical character recognition reader and a radio frequency signal identifier reader for reading the unique radio frequency signal identifier from the radio frequency emitting tag may be included on a single computer. The associating the unique radio frequency signal identifier with the unique electronically formatted waste container identifier step may

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be performed by a single computer. The method may further comprise the step of transmitting the unique radio frequency signal identifier and the unique electronically formatted waste container identifier associated therewith to a central computer. The transmitting step may be performed by a single computer. The method may further comprise the step of distributing the waste container having the unique optical waste container identifier pre-printed thereon to the user of the waste container.

A second aspect of the present invention is directed to a method of deploying multiple waste containers in a waste collection system. The method comprises the steps of: (1) providing an optical character recognition reader capable of reading a unique optical waste container identifier printed on a waste container and converting the unique optical waste container identifier to a unique electronically formatted waste container identifier; (2) providing a radio frequency emitting tag comprising a unique radio frequency signal identifier stored thereon; (3) providing a software for associating the unique electronically formatted waste container identifier with the radio frequency emitting tag identifier; (4) providing a software module for associating the radio frequency signal identifier with personal information data associated with a user of the waste container; and (5) providing a software module for compiling a database comprising the personal information data associated with the user of the waste container, the unique radio frequency signal identifier, and unique electronically formatted waste container identifier.

The second aspect of the invention may include one or more of the following features or steps, alone or in any reasonable combination. The method may further comprise the step of providing a radio frequency emitting tag reader for reading the radio frequency signal identifier on the radio frequency emitting tag. The method may further comprise the step of providing a computer wherein the radio frequency emitting tag reader and the optical character recognition reader are located on the computer. The software for associating the unique electronically formatted waste container identifier with the radio frequency emitting tag identifier may be stored in a memory located on the computer. The software for associating the radio frequency signal identifier with personal information data associated with the user of the waste container may be stored in the memory located on the computer.

Other features and advantages of the invention will be apparent from the following specification taken in conjunction with the following drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

To understand the present invention, it will now be described by way of example, with reference to the accompanying drawings in which:

FIG. 1 is a perspective view of a waste removal and hauling vehicle of the present invention;

FIG. 2 is a perspective view of the vehicle of FIG. 1 with a waste receptacle in an unloading position;

FIG. 3 is a perspective view of a second waste removal and hauling vehicle of the present invention;

FIG. 4 is a perspective view of the vehicle of FIG. 3 with a waste receptacle in a second position;

FIG. 5 is a perspective view of a third waste removal and hauling vehicle of the present invention;

FIG. 6 is a perspective view of the vehicle of FIG. 5 with a waste receptacle in a second position;

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FIG. 7 is a perspective view of a fourth waste removal and hauling vehicle of the present invention;

FIG. 8 is a perspective view of the vehicle of FIG. 7 with a waste receptacle in a second position;

FIG. 9 is a front and rear perspective views of a receptacle showing placement of a communication means;

FIG. 10 is a flowchart of a method of the present invention;

FIG. 11 is a flowchart of a process of an on-board computer;

FIG. 12 is a flowchart of a process for controlling automatic receptacle emptying carried out by an on-board computer;

FIG. 13 is a flowchart of a process of a first receiver located at a first position on a waste removal and hauling vehicle;

FIG. 14 is a flowchart of a process of a second receiver located at a second position on a waste removal and hauling vehicle;

FIG. 15 is a flowchart of a communication process carried out by an on-board computer; and

FIG. 16 is a block diagram illustrating an apparatus and a method of the present invention;

FIG. 17 is a flowchart illustrating a method by which a municipality may use principles and concepts of the present invention to control, track, and monitor waste or refuse receptacles

FIG. 18 is a block diagram of an aspect of the invention for use in conjunction with the method FIG. 17; and

FIG. 19 is a block diagram of an aspect of the invention for use with the method of FIG. 17

FIG. 20 is a schematic showing a waste container and a handheld computer wherein the waste container has a unique alphanumeric identifier and a radio frequency tag attached thereto having a unique radio frequency identifier stored thereon, and the handheld computer has an optical character recognition reader, a radio frequent tag reader, a memory with executable software stored thereon, and a means for communicating with a remote server or computer;

FIG. 21 is a schematic showing a waste container and a handheld computer wherein the waste container has a unique alphanumeric identifier and a radio frequency tag attached thereto having a unique radio frequency identifier stored thereon, and the handheld computer has an optical character recognition reader, a radio frequent tag reader, a memory with executable software stored thereon, and a means for communicating with a remote server or computer; and

FIG. 22 is a schematic diagram of the handheld computer of FIGS. 20 and 21 in communication with a central server.

DETAILED DESCRIPTION

While this invention is susceptible of embodiments in many different forms, there is shown in the drawings and will herein be described in detail preferred embodiments of the invention with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the broad aspect of the invention to the embodiments illustrated.

Referring to FIGS. 1-16, waste removal and hauling vehicles and systems or methods incorporating such vehicles with sensors and an operational network to facilitate proper, efficient and effective waste monitoring and collection are illustrated. As best illustrated by the block diagram in FIG. 16, the overall system 10 comprises a receptacle 100, preferably a plurality of receptacles located at a plurality of locations throughout a municipality, a refuse or waste haul-

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ing vehicle 200, an external site 300, typically a central station, management office, dispatcher or the like, and a handheld reader/prINTER transmitter means 400 used in an example of a municipal functionality.

Each receptacle 100 is outfitted with a means of communicating or transmitting a signal carrying an identifier associated with the receptacle 102, such as a radio frequency identification (RFID) tag. RFID tags are well-known in the retail industry for identifying products. In particular, an RFID tag or transponder is an object applied to or incorporated into a product for the purpose of identification using radio waves. Most RFID tags contain at least two parts, namely an integrated circuit for storing and processing information, modulating and demodulating a (RF) signal and an antenna for receiving and transmitting the signal. RFID tags come in three general varieties: passive, active, or semi-passive (also known as battery-assisted). Passive tags require no internal power source, thus being pure passive devices (they are only activated when a reader is nearby to power them), whereas semi-passive and active tags require a power source, usually a small battery. While the present discussion focuses on passive RFID tags, it is recognized that a similar system can be used for active and semi-active RFID tags.

The RFID tags may transmit high frequency (HF) signals, low frequency signals (LF), or ultra-high frequency (UHF) signals. The characteristics of these signals differ, and one particular frequency may be more advantageous than another based on the particular application of the system. For instance, low frequency signals have a short read range (distance) which would prevent errors or false reads. UHF signals can have a read range that is very short to very wide (any yards or meter).

While RFID, and in particular passive RFID tags are discussed herein, it is recognized that other identifying or alerting tags can be used. Many are known and perhaps others in the future. For example, electrical article surveillance (EAS) tags can be used. EAS tags are commonly used to prevent shoplifting from retail stores. The EAS tags are attached to merchandise and are removed or deactivated by the clerks when the item is properly bought or checked out. At the exits of stores, a detection system sounds an alarm or otherwise alerts the staff when it senses active tags. These tags are typically made of a strip of amorphous metal (metglas) having a low magnetic saturation value. This strip is also lined with a strip of ferromagnetic material with a coercive field (magnetic "hardness"). Detection is achieved by sensing harmonics and sum or difference signals generated by the non-linear magnetic response of the material under a mixture of low-frequency (in the 10 Hz to 1000 Hz range) magnetic fields.

When the ferromagnetic material is magnetized, it biases the amorphous metal strip into saturation, where it no longer produces harmonics. The tags are activated by demagnetization and deactivated with magnetization.

The identifier associated with the receptacle is preferably a discreet identifier which is assigned to the receptacle 100. The identifier information is stored on a database typically located at the external site 300, and electronically joined with an account to which the receptacle 100 belongs. In other words, account information is housed on a database located at the external site 300. Each account has one or more receptacle identifiers associated with it, and the database carries with it information typical to the management of any business account, for example, special instructions, accounts receivable, last receipt, last invoice, amount in

arrears, days since last payment, historical account information, contact information, owner, etc.

Referring to FIG. 1-8, various vehicles **200a-d** used in conjunction with the system **10** are illustrated. These vehicles **200a-d** are intended to be examples of the types of waste removing and hauling vehicles used to transport refuse today. It should also be understood that the vehicles **200a-d** and related concepts of the external site **300** and the receptacle **100** can be used alone or, more importantly, in combination with the municipal permitting concepts described above

Each of the vehicles **200a-d** includes a waste bin **202** located above a baseline upon which the vehicle **200a-d** is supported, generally the ground. The waste bin **202** includes a chamber **204** and an emptying site **206**. The refuse within the receptacles **100** is loaded into the chamber **204** via the emptying site **206**. One of ordinary skill in the art of waste hauling would readily understand this method of refuse handling without further description as it is the standard procedure employed in the art.

The vehicles **200a-d** are further outfitted with at least one receiver **208a**, in most cases a plurality of receivers **208a,b** and preferably two. The receivers **208a,b** may be sensors, transducers, or antennae, or any combination thereof. As illustrated in FIGS. 13 and 14, preferably, the receivers **208** receive the signal from the transmitting means **102** on the receptacles **100** and send a corresponding signal via wire, wireless, or any other medium to an on-board communication or data link **210**. The signal from the receiver **208a,b** may be response signal to the energy in the form of the signal transmitted by the transmitter means **102**, or the signal may simply be a pass through signal, including conversion from a wireless signal to a signal carried by another medium such as a wire. In either case, all or a part of the signal from the receivers **208** to the data link are associated with the actual identity or the unique identifier associated with the each receptacle **100**. In the case where a separate response signal to the transmitter means **102** is generated, the response signal may carry additional information such as location of the particular receiver **208a,b** generating the signal, etc.

The locations chosen for the receivers **208a,b** are extremely important because the receivers **208a,b** are used for several different purposes, including for possible tracking of the receptacle **100** as it is transferred from position-to-position relative to the vehicle **200a-d** by a means for transferring the receptacle **212**, which may be automated systems known in the art, as shown in FIGS. 1-8, comprising mechanical arms, booms, or the like extending outwardly from the vehicle **200** to which a waste receptacle **100**, or a waste bin **202** in the case of a roll-off vehicle **200d** or manual systems also known in the art. For instance, in a conventional refuse collection and hauling vehicles **200a-d** illustrated in FIGS. 1-8, a first receiver **208a** is located at a position where reception of the signal from the transmitter means **102** can be promoted prior to unloading or emptying of the receptacle. As shown in the FIGS. 1-8, the first location or position is typically where the receptacle **100** is engaged by the means for transferring the receptacle **212** prior to the receptacle **100** being transported to the emptying site **206** by the means for transferring **212**.

The preferred location of the first receiver **208a** on a front loading vehicle **200a**, illustrated in FIGS. 1 and 2, is on the front of the vehicle **200a** below the windshield and centered between opposing ends or sides. The first receiver **208** may be recessed or spaced from the bumper such that accidental collision with an obstacle would not damage the receiver.

Accordingly, the first receiver **208a** may be located within or protected by the shell of the vehicle **200a** as long as the receiver **208a** is capable of achieving its purpose. One example of such a location may be behind the vehicle radiator grill. In any event, the first receiver **208a** is associated with the means for transferring the receptacle **212** when the receptacle is in a first position. As illustrated, the means for transferring **212** comprises a pair of arms with grasping hooks for engaging corresponding handles on the receptacle **100**.

FIGS. 3-6 show vehicles emptying a certain type of receptacle **100**. This type of receptacle **100** is illustrated in FIG. 9 to show a placement position for the communication means **102** which may not be evident from FIGS. 3-6.

The preferred location for the first receiver **208a** on a side loading vehicle **200b**, illustrated in FIGS. 3 and 4, is on a side of the vehicle corresponding to a side of the vehicle where the receptacle **100** would be located prior to or just as the means for transferring the receptacle **212** engages the receptacle **100**. This location is typically on the passenger side adjacent the transfer means **212**. This includes either just behind the cab or on the cab to the rear of the passenger side door. The first receiver **208a** may be protected by locating on the passenger side and behind the cab, such that the receiver is protected by the cab and the waste bin **202**.

The preferred location of the first receiver **208a** on a rear loading vehicle **200c**, illustrated in FIGS. 5 and 6, is at the rear of the vehicle **200c** above the baseline, between the tail lights, but below the emptying site **206**. Thus, as receptacles **100** are transported to the vehicle **200c** for emptying at the emptying site **206**, the communication means **102** may be positioned such that the first receiver **208a** may receive the signal from the communication means **102** without difficulty. These types of vehicles **200c** may or may not be outfitted with an automated means for transferring receptacle **212**. The example shown in the drawings exhibits such an automated system.

The preferred location of the first receiver **208a** on a roll-off vehicle **200d**, illustrate in FIGS. 7 and 8, is similar to the location on rear loading vehicles **200c**, namely at the rear of the vehicle **200d**, above the baseline and between the tail lights of the vehicle **200d** where the means for transferring **212** first engages the waste bin **202**. Although one of ordinary skill in the art would readily recognize this point, it should be noted that a roll-off vehicle **200d** differs from the other vehicles **200a-c** in that the waste bin **202** also acts as the receptacle. Thus, the transfer means **212** does not transfer and dump a receptacle. Rather, it engages a waste bin **202** and loads it onto the vehicle **200d** bed. It follows that the transmitter means **102** would be located on the waste bin **202**, preferably on the forward or front facing wall as shown in the drawings.

The vehicles **200a-d** also include a second receiver **208b** located at a position where reception of the signal from the transmitter means **102** can be promoted either at the instant of receptacle unloading or just prior or just subsequent to receptacle unloading. Accordingly, this second location is typically adjacent the emptying site **206** to the chamber **204** and in alignment with the first receiver. The phrases "in alignment with the first receiver **208a**," "aligned with the first receiver **208a**," and the like are intended to include a position wherein a path of the receptacle and the means for transmitting **102** taken from the first position at the first receiver **208a** to a second position at the second receiver **208b**, as determined by the means for transferring the receptacle **212** included on the vehicles **200a-c**, naturally brings the means for transmitting **102** to a location where the

second receiver **208b** can receive the signal from the means for transmitting **102**, i.e. into a range of the second receiver **208b** where reception is achieved.

To further facilitate accuracy in the system **10**, a means for activating **213** the first and/or second receivers **208a,b** may be included in the system **10**. For example, the second receiver **208b** may be activated or powered up only upon waste receptacle **212** reaching a certain point in the process of transferring the receptacle **212** from the ground to the emptying site **206**. For example, a proximity switch, photo sensor, laser, mechanical switch or the like can be employed to trigger the second receiver **208b** such that the receiver **208b** can receive a signal from the transmitter means **102**. This improvement reduces the number of false reading that could be taken by the receivers **208a,b** during the receptacle **212** emptying process.

In one embodiment, a means for activating **213** signals the first receiver **208a** to activate upon grasping of the waste receptacle **100** by the automated means for transferring the receptacle **212**. The means for activating **213** signals the first receiver **208a** to power down when the waste receptacle **100** is mover upwardly. A second means for activating **213** is associated with the second receiver **208b** and signals the second receiver to power up when the means for transferring **212** reaches a predetermined location, generally a height, with the receptacle **100**. Once the reading is taken by the second receiver **208b**, the means for activating **213** signals the second receiver **208b** to power down. Thus, the first and second receivers **208a,b** are only able to receive signals from an RFID tag of a given receptacle **100** when the means for activating **212** allows the receivers **208a,b** to be powered up. This greatly eliminates false readings taken by the first and second receivers **208a,b**.

In one embodiment, the means for activating is triggered or activates the first or second receiver **208a,b** based on the position and/or movement of the means for transferring **212** through a given zone, region, or area located on the truck, preferably long a path taken by the receptacle by the means for transferring or by a path taken by the means for transferring **212** itself.

The preferred location of the second receiver **208b** on a front loading vehicle **200a**, illustrated in FIGS. **1** and **2**, is on the front of the vehicle **200a** above the windshield and preferably above the cab, approximately aligned with the first receiver **208a**. The second receiver **208b** may be attached to the roof of the cab or closer to the emptying site **206**, which is an opening on the vertically upper surface of the bin **202**, on a portion of the bin **202** just behind the cab and aligned with the first receiver **208a**. The second receiver **208b** may be located within or protected by the shell of the vehicle **200a** or the bin **202** as long as the receiver **208b** is capable of achieving its purpose.

The preferred location for the second receiver **208b** on a side loading vehicle **200b**, illustrated in FIGS. **3** and **4**, is on a side of the vehicle corresponding to a side of the vehicle where the receptacle **100** would be located and elevated above the position of the first receiver **208a**. This location is typically on the passenger side either just behind the cab or on the cab to the rear of the passenger side door, either on the roof of the cab, on the rear of the cab adjacent the roof, or on some other portion of the vehicle **200b**, such as an upper, stationary portion of the transfer means **212** as illustrated. In any case, the second receiver **208b** is aligned with the first receiver **208a** as determined by the path of the receptacle **100** provides by the means for transferring the receptacle **212**.

The preferred location of the second receiver **208b** on a rear loading vehicle **200c**, illustrated in FIGS. **5** and **6**, is at the rear of the vehicle **200c** above the baseline, between the tail lights, and above the first receiver site **208a** and aligned therewith.

The preferred location of the second receiver **208b** on a roll-off vehicle **200d**, illustrate in FIGS. **7** and **8**, is generally some portion of the vehicle **200d** adjacent to where the front face of the bin **202** having the transmitter means **102** is located once the bin **202** is loaded onto the vehicle **200d**. This may be on the back of the cab adjacent to where the means for transmitting **102** on the waste bin **202** is positioned once the transfer means **212** positions the waste bin **202** on the vehicle **200d**, or it may be on a forward portion of transfer means **212** as illustrated. Again, the second receiver **208b** is preferably aligned with the first receiver **208a**.

As shown in the block diagram of FIG. **16**, the vehicle **200** has a controller **214**, for activating and deactivating the means for transferring the receptacle **212**, in communication with an on-board computer **216**. The on-board computer **216** has a graphic interface **218** and provides the data or communication link to the external site **300** via modem or some other communication device. The controller **214** may also be a module within the on-board computer without departing from the spirit of the invention. Communication to and from the external site **300** is shown as wireless communication. The first and second receivers **208a,b** are illustrated as spaced wherein as the transmitting means **102** traverses toward the first receiver **208a**, the first receiver **208a** receives the signal from the transmitter means **102**. Likewise, as the receptacle **100** is transferred to the second position where the second receiver **208b** is located, the second receiver **208b** receives the signal from the transmitter means **102**. Movement of the receptacle **100** and the associated signal are shown in dashed lines.

The external site **300** may include a server **302** in communication with computer **304** and a database **306**, typically on the computer **304**. Of course, the server **300** is not required to be at the same physical site as the computer **304**, nor is it required for the database **306** to be stored on a computer separate from the server **302**. The block diagram is merely an example of a possible layout. The only requirement for the external site **300** is the database **306** and a means for communication between the vehicles and database **306**.

Now, as illustrated in flowcharts of FIG. **10-15**, the various components of the system, the receptacle **100**, the vehicle **200**, and the external site **300**, work together to provide an overall apparatus, the elements of which were described above, and a method for removing and hauling waste.

Referring to FIG. **10**, first, a receptacle **100** is located at a baseline position adjacent the vehicle **200** such that means for transmitting **102** is brought within the reception range with the first receiver **208a**. The transmitter means **208a** sends a signal carrying discreet receptacle identifier information to the receiver **208a**. The receiver **208a** either passes the signal directly to the on-board communication link, preferably a module of the on-board computer **216**, or generates a new signal based on the signal received from the means for transmitting, but still including some type of receptacle identifier.

Next, the communication link sends a signal, either pass through or new, to the external site which receives the signal. The identity of the receptacle **100** is checked against an account database **306** to verify that the receptacle **100** should

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be emptied into the vehicle 200. A signal is generated indicating the account status associated with the receptacle identifier.

If the account is in good standing (pass condition), the controller 214 for the means for transferring the receptacle 212 is automatically activated either by a module or routine on the on-board computer or on the external computer 304. The means for transferring the receptacle 212 transports the receptacle 100 to the emptying site 20 where the transmitter means 102 is within reception range of the second receiver 208b. The transmitter means 102 sends a signal carrying discreet receptacle identifier information to the receiver 208b. The receiver 208b either passes the signal directly to the on-board communication link, preferably a module of the on-board computer 216, or generates a new signal based on the signal received from the means for transmitting, but still including some type of receptacle identifier. The triggered or pass through signal from the second receiver 208b represents an event. The event is preferably the emptying of the receptacle 100 into the vehicle bin 202. This event is recorded on the database 306 at the external site 300 and associated with the receptacle account.

If the account is not in good standing (fail condition), the controller 214 is not activated, a signal carrying an alarm or warning is transmitted to the vehicle 200. The account status can also be displayed on the on-board graphic interface 218. As a result, the transfer means 212 is not activated, and the receptacle 100 is not emptied into the vehicle bin 202. However, if for some reason, such as immediate, on-the-spot payment of an invoice by a customer, the vehicle personnel are inclined to empty the receptacle 100, vehicle personnel may override the alert and manually engage/energize the transfer means 212. In this case, as illustrated on the right hand side of FIG. 10, transmitter means 102 sends a signal carrying discreet receptacle identifier information to the second receiver 208b. The second receiver 208b either passes the signal directly to the on-board communication link, preferably a module of the on-board computer 216, or generates a new signal based on the signal received from the means for transmitting, but still including some type of receptacle identifier. The triggered or pass through signal from the second receiver 208b represents an event. The event is preferably the emptying of the receptacle 100 into the vehicle bin 202. This event is recorded on the database 306 at the external site 300 and associated with the receptacle account. The first receiver 208a may send an additional signal at a second moment in time when the receptacle 100 at is returned to the first position.

Referring specifically to FIG. 11, a flowchart of a process of the on-board computer 216 is illustrated. Here, the communication or data link between the receivers 208a,b and the external site 300 is a module or routine provided by the computer 216. This process comprises receiving a signal carrying the receptacle identifier from the first receiver 208a. The computer 216 then transmits a signal from the vehicle 200 to the external site 300. Once the data has been analyzed at the external site 300, the computer 216 receives a return signal from the external site 300 and determines whether to activate the transfer means 212 or require an override to empty the receptacle 100. If the transfer means 212 is activated, either automatically or manually via override, the second receiver 208b will send a signal to the computer 216. The computer 216 transmits a second signal to the external site 300 indicating that the emptying event has taken place.

Referring specifically to FIG. 12, a process for controlling the transfer means 212 is illustrated. This process may be performed by the on-board computer 216 or by an existing

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controller 214 in communication with the on-board computer 216. The computer 216 receives a signal related to or determined by an account status associated with the receptacle. If the account status is such that the receptacle should be emptied, i.e. current, a signal is generated to automatically energize the transfer means 212. If the account status is not current, the process ends.

Referring to FIG. 15, a process of the computer 304 of the external site 300 is illustrated. The main purpose of the server 302 and/or the computer 304 of the external site 300 is to hold or have the capability of accessing a database 306, the database 306 including an association between receptacle identity, ownership or responsibility, and account status. In terms of the overall system or method of the invention, the external site 300 communicates with the vehicle 200 by receiving and sending signals.

For example, the external site 300 receives a first signal carrying a receptacle identifier from the vehicle 200. This first signal is associated with the receptacle 100 being located at the first position. The first signal originates with, though is not necessarily identical to, the signal received from the transmitter means 102 at the first receiver 208a. A module at the external site 300, preferably on the computer 304, compares the data carried by the signal against account information on the database 306 to determine an account status associated with the receptacle identifier. The result of the comparison, i.e. the account status, is transmitted from the external site 300 back to the vehicle.

Further, the external site 300 receives a second signal from the vehicle 200. This second signal is associated with the receptacle 100 being located at the second position. The second signal originates with, though is not necessarily identical to, the signal received from the transmitter means 102 at second receiver 208b. A module at the external site 300, preferably on the computer 304, compares the data carried by the signal against account information on the database 306 and records an event, the emptying of the receptacle 100, in connection with the receptacle 100.

Referring generally to FIGS. 17-19, the inventors also foresee the means for communicating or transmitting 102 as being supplied by municipalities on a permit basis to citizens. This would provide several advantages. First, the permit process would provide an unexpected source of income to the municipality because the typical permit granting process comes with an associated government fee. Secondly, it would provide a means by which the municipality can easily track ownership of or responsibility for the receptacle. Thirdly, because ownership of the receptacle would be stored on the database created during the permit process, citations for misuse, neglect, damage, etc. can be easily, swiftly, and electronically generated using a handheld receiver with printer for generating citations, providing yet a further stream of income to municipalities as well as improving lane conditions, thus reducing the likelihood of infestation by vermin. In addition, sanitation inspectors would be more productive due to the automation provided by the handhelds readers in combination with the communication means, preferably RFID tags 102.

As set forth above, this aspect of the invention directly results in cleaner streets and alleys. The invention will eliminate or reduce trash overages, under size containers, poor container maintenance condition, e.g. no lid. Permits and citations will force compliance. Users can specify correct container size, schedule additional pick-ups. Service may be halted due to lack of payment or by schedule.

Information regarding each permit/receptacle identifier, the account associated therewith, and the entity responsible

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for the account/permit/receptacle is stored on a managed database. The database may include other information such as hauler name, permit number and container asset number (human readable sticker and RFID tag). Use and access of the database is explained in more detail below.

The RFID tags **102** can be read by inspectors having mobile and/or handheld computers **400**. Citations may be issued immediately via printer on the handheld **400** and/or mailed with back-up violation data. Inspectors and other municipal employees use mobile handheld RFID readers with cameras to read tags and report violations. Citations can be issued and wireless transmission of data achieved through the handhelds **400**.

This aspect of the invention requires haulers or receptacle owners to purchase annual container permits. The issued permit includes a means for transmitting **102**, such as an RFID container tag. Revenue to the municipality is generated by an enforcement program.

Further, a container registration fee includes RFID tagging. The RFID tag information is linked to customer and hauler information in the database.

More specifically, a tag **102** (such as a passive RFID tag) acts as a permit. While the description below discusses passive RFID tags as permits, it is recognized that other information storing devices can also be used, such as active RFID tags, hybrid RFID tags, smartcards, and other hereto unknown items.

Typically, one goes to the governing body or agency and fills out the necessary registration forms, pays the necessary fees, and receives the RFID tag **102** acting as a permit. This initial registration and payment can also be done on-line through the internet or over the telephone. The tag includes certain necessary information dependent on its purpose. Encoding, entering, and keeping such information on the tag is well-known to those skilled in the art. Such information can include: the permit's issuing body (e.g., Secretary of State, Village of Anywhere, City of Nowhere), the date of

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issuance of the permit, the term of the permit (e.g., Jun. 1, 2010-May 31, 2011), the type of permit (e.g., "Permit for Trash," "Vehicle Permit," Elevator Permit), the location for the permit's use (e.g., "123 Main St. Anytown, USA," "1999 Ford Austere," "Elevator No. 6 SE for The Towers"), the responsible party's name and contact information (e.g., "Joe Smith at 111 S. Main, Anytown," "Office of Management, The Towers, 432 E. Main, Anytown," "Joe's Construction Co., 999 W. Main, Anytown"), and any other information necessary under the circumstances for the inspector to make a determination as to whether the permit is in compliance or out of compliance resulting in a violation. In the preferred application, the tag **102** includes all the information necessary for a government or independent inspector to make a determination in the field at a remote location as to whether a violation is present and has occurred relating to the item being inspected.

For visual inspection and to ensure proper location (e.g., the tag is in the correct vehicle or at the correct address), the RFID tag **102** typically has identifying indicia thereon such as time period involved, address, municipality's identity, and possibly the owner's identity. This ensures the tag **102** is placed at the proper location and gives the inspector visual verification of the tag's location.

The tag **102** is given or sent to the person responsible for the permit along with specific instructions as to placement, use, and purpose.

The **102** tag can also include a mounting component **104**, such as a protective pocket or pouch having an adhesive on one side so the pouch can hold the tag and protect it and be adhered to the location where the permit is to be displayed, e.g., the outside wall of a recycling container, the inside of a trash can lid (outside the trash can itself) or on the outside can wall, the inside of a windshield, the inside of a business window, etc.

EXAMPLES

Permit Purpose	Permit	Identification Information	Carrier	Location
Trash/Refuse Removal	The Collection, Storage of Trash/Refuse for Removal and Size of Container, Compactor or Bailer	Address Type of Container Permit Number & Date Term/Period Valid	Protective Pocket with adhesive on one side	Placed on inside of lid for Trash Container or outside Large Compactor/Bailer
Recyclables Removal	Collection of The Collection, Storage and Removal of Recyclable Materials and Size of Container, Compactor or Bailer	Address Type of Container Permit Number & Date Term/Period Valid	Protective Pocket with adhesive on one side	Placed on inside of Lid for Recyclable Container or outside Large Compactor/Bailer
Grease Boxes	The Collection and Storage of Discarded Grease for Removal	Address Type of Container Permit Number & Date Term/Period Valid	Protective Pocket with adhesive on one side	Placed on inside of Lid for Recyclable Container or outside of Box

-continued

Permit Purpose	Permit	Identification Information	Carrier	Location
Business	To Conduct Business at a Certain Premise	Address City Name Permit Number & Date Term/Period Valid	Protective Pocket with adhesive on one side	Placed on inside of Window for Business
Vehicle	To Park a Vehicle Residing within Municipality's Borders	Type of Vehicle City Name Permit Number & Date Term/Period Valid	Protective Pocket with adhesive on one side	Placed on inside of Window for Vehicle
Fuel Dispensing	To Dispense Accurate (Inspected) Amounts of Fuel within Municipality's Borders	Address City Name Permit Number & Date Term/Period Valid Issuing Body and Contact Info.	None	Placed on Dispensing Mechanism
Elevator	To Safely Operate (Inspected) Elevator within Municipality's Borders	Address City Name Permit Number & Date Term/Period Valid Issuing Body and Contact Info.	Frame	Placed in Frame on Elevator Wall
Construction	To Perform Construction (New, Addition, Improvement) within Municipality's Borders	Address City Name Owner Contractor Permit Number & Date Term/Period Valid Issuing Body and Contact Info.	Protective Pocket with adhesive on one side	Placed at Construction Site (In Window or on Sign Advising of Construction)
Billboard	To Display Public Advertisement	Location City Name Owner Permit Number & Date Term/Period Valid Issuing Body and Contact Info.	Protective Pocket with adhesive on one side	Place at Bottom or on Back of Billboard Stand
Portable Toilets	To Safely Operate a Healthy Facility within Municipality's Borders	Location City Name Service Provider-Owner User/Customer Permit Number & Date Term/Period Valid Issuing Body and Contact Info.	Protective Pocket with adhesive on one side	Placed inside Unit
Newspaper Stands	To Conduct Business at a Certain Location	Address City Name Permit Number & Date Term/Period Valid	Protective Pocket with adhesive on one side	Placed on Conspicuous Location on Stand

The handheld device **400** is preferably a small computer carried by an authorized remote user. It is the method used to gather information in the field by the inspector, such as a policeman, village employee, etc. The device **400** includes several components, that being a power source **401**, such as

a battery, a means to enter data (a reader and keyboard) **402**, an internal clock **404**, a memory **406**, a GPS locator **408**, a camera **410**, a monitor or screen **412**, and a printer **414** or preferably a wireless connection **416** to a printer **414** having a wireless communication means **418**.

One or more methods can be employed to enter the necessary data into the remote handheld device **400**. The preferred means is an RFID reader **420**. The reader **420** is capable of reading passive RFID tags **102** in the field. A keyboard **412** may also be provided for entering additional data as may be necessary.

In particular, three items are inputted into the remote handheld device **400**, that being an authorization code, the permit information, and the time and date. The user's authorization code and the date and time can be entered on a keyboard provided on the handheld device **400**. However, the preferred method of entering such data is by a reader **420** associated with the handheld device **400**. The reader **420** can read the tags **102**, such as passive RFID tags. The user/inspector can thus have an identification RFID card with the necessary authorization codes thereon to activate the device **400** and to operate and use the device **400** once the reader **420** reads the authorization information. Similarly, the reader **420** can read the information provided on the permit, such as a license plate number, a home address, a business address, etc. The date and time can be entered automatically by an internal clock **404** in the device **400**.

A internal clock **404** is provided to document the time certain events occur, such as the reading, gathering and/or entering of information associated with reading the permit, the date/time of a photograph, and the date/time a citation is entered or printed, etc. The clock **404** is an internal mechanism stamping the memory of the event (permit reading, photograph, GPS locating, citation issuance) with the date and time of the event.

The device **400** includes a screen or monitor **412** for displaying information read and received. The device **400** also includes a wireless connection or link (Bluetooth) **416** to a mobile printer **414**. The device **400** can then display or print information obtained by the reader **420** in the device **400** relating to the tag **102** or generated by the device **400** and relating to a citation. In the alternative, the printer **414** need not be mobile, but can be used at a central location **500** so that when the device **400** is downloaded at a central location **500** (such as the police station or city hall) through conventional means, information read by it or citations generated by it can be printed out.

An optional camera **410** and/or GPS locator **408** may also be provided in the handheld device **400** to document both what is seen visually by the remote user and to identify the location of the remote user.

Once the data has been entered or citation processed, the entry or issuance can be confirmed by digital or physical signature. The device **400** can have a touch screen for recording a signature by the inspector. In the alternative, the inspector's authorization card can be read once more or drawn from the device's memory **406** to confirm the inspector's identity and authorization relating to the entry of information or issuance of a citation.

As noted, biographically and permit information is obtained from an applicant for the permit and stored in the tag **102**. For trash, refuse and recyclables this information will also include time period of permit (e.g., 6 months, 1 year, etc.), number of containers, location of containers (address), type of containers (e.g., size), and hauler information (identity, contact information, pick-up schedule, etc.). A permit, in this case—an RFID tag **102**—is issued by the authorized body. The RFID tag **102** will typically have identifying indicia thereon such as address of containers, and possibly types and/or size of containers. The RFID tag **102** has an adhesive thereon or is placed into a protective pouch (having a transparent cover, with an adhesive on an

outwardly facing surface). The RFID tag **102** is then adhered to the outer container wall in a designated location.

A remote, authorized user, typically a government or third party inspector enters an authorization code by typing it in or reading it (or it is automatically drawn from memory **406** and the initial sign-on/authorization), on her handheld device **400**. When at the remote location, the inspector then reads the RFID tag **102** (the permit information contained thereon) with the handheld device **400**. The display **412** (or pint-out) will then display the necessary information for the inspector to make a determination as to whether the permit is in compliance or whether there is a violation. Such violations can include, among others, the permit is outdated and expired, the containers **100** of the permit are not the same as those being used, the containers **100** are constantly and consistently overflowed meaning they are undersized, the containers **100** are damaged and have not been properly maintained, the containers **100** are missing lids/covers, the containers **100** are in an improper location, the containers **100** do not have a permit, and/or the hauler has not made the necessary and scheduled pick-ups of recyclables and/or refuse. Of course, one of the primary purposes of issuing citations is to force compliance with rules, regulations and ordinances to maintain properly sized, safe equipment.

If desired, to document or corroborate the violation, the inspector can take a photograph of the scene with a camera **410** built-into or associated with the handheld device **400** and obtain the GPS location of the scene with a GPS locate **408** built-into or associated with the handheld device **400**. This documentation stored in memory **406** can accompany the citation or merely be kept in memory **406** (on the handheld device or at a central database **502** at the central station **500**) as back-up. It is preferred to have such documentation accompany and be part of the citation.

Once a violation has been determined, the inspector can issue or process a citation. This can be accomplished many ways. One way is to use the information read from the tag **102** and in the memory **406** of the handheld device **400** and entering the necessary information associated with the violation. The citation can then be stored in the memory **406** of the device **400** and formally issued later via mail or similar once the inspector's tour is completed. In the alternative, the device **400** can include a wireless link **416** to a mobile printer **414** for printing the citation and giving it or leaving it with the responsible party (business owner, individual, etc.). The violation and citation records can further be stored in the device's memory **406** for downloading later for documenting the citation, record keeping, tracking and follow-up. If the device **400** was used to take a picture and/or obtain location information, that information can also be attached or used with the issued citation as confirmatory information and supporting documentation.

It should be recognized that while the handheld device **400** is discussed as being carried by an inspector, it can also be mounted on a vehicle. As a result, the tag-permit can also be read by an RFID or similar reader mounted to a mobile vehicle. In some instances, this may prove to be more efficient.

The above describes a system wherein the tag-permit **102** and handheld device **400** are the primary components and virtually the sole components of the system. In an alternative device, the device **400** includes a communications module **416** to communicate with a central station **500** having a central database **502** and the tag-permit **102** has encoded information thereon for directing the central station **500** to make a determination of whether there is a violation and a citation should be issued.

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Briefly, while the tag **102** acts as the permit, it merely includes identifying information encoded therein. As before, it may also have identifying indicia thereon such as time period involved, address and municipality's identity and possibly the owner's identity. Again, a mounting component **104**, such as a pouch, can be used also.

The registration information, instead of being stored on the permit, is stored in a memory **504** associated with the central station **500**. The central database **502** is housed and maintained at the central location **500** such as a police station, village hall, county office or state agency, or third party service provider. Some, if not all of the information is obtainable by and from an authorized remote user. The central database **502** is maintained on a system, such as on a computer **506**, that can receive queries and transmit select information automatically or by pointed requests. As a result and by way of example, a policeman may access the database **502** and request certain vehicle and owner information by entering a license plate number or by specific inquiry prompted by reading the RFID-permit.

A communication module **508** is associated with the central station **500** allowing data or information to be received from one or more remote handheld devices **400** and information to be transmitted to one or more remote handheld devices **400**.

Housed with the central station **500** is look-up and/or comparison software **510** capable of receiving information from the handheld device **400**, polling/querying the memory **504** associated with the central station **500**, and if necessary, making a comparison and generating a report or result of the comparison made. The information polled/looked up or generated by a comparison is then sent to the remote handheld device **400**.

Specifically, the information obtained from the handheld device **400**, such as a license plate entered into the remote handheld device **400** or by reading the tag **102** with an RFID reader **420**, and received by the central station **500** will be looked-up or compared with the information accessed from the central database **502**. The individual operating the remote device **400** can then determine whether a violation has occurred (such as when the looked-up data shows the permit has expired). Or, the database **502** comparison will then determine if the information obtained is the same as the information stored and/or determine whether a violation has occurred, such as the permit period paid for has lapsed.

The handheld device **400** is described above and is similar, except it further includes a communications module **416** for transmitting and receiving communications from the central station **500**.

As noted, biographically and permit information is obtained from an applicant and stored in memory in a central database **502**. A remote, authorized user, typically a policeman, government employee, or third party or government inspector, once authorized, reads the RFID-permit by the handheld device **400**. This information is transmitted to the central database **502** where, in turn, the database queries its memory for information necessary to determine if a violation occurred, such as the permits expiration or improper container size. A citation can then be issued. And printed at the handheld device **400** or on a printer **512** at the central station **500**.

By way of example, with respect to vehicle stickers, a traffic control manager can read an RFID attached to a windshield and transmit the information to a central database **502**. The central database **502** then polls its memory **506** for information associated with that permit. The database or the manager in the field can then determine if the

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permit is expired or if the permit is associated with the correct vehicle. If a violation has occurred, the manager can issue a citation by printing a citation and attaching it to the car or transmitting the necessary information to the central database for printing and issuing (by mail) the citation.

Referring to FIGS. **20-22**, a method of deploying or redeploying multiple waste containers **100** in a waste collection system as described above. The general purpose of the method is to outfit waste containers **100** with radio frequency tags **102** in a manner that overcomes the drawbacks of current solutions.

Typically, waste containers **100** are produced with a unique optical alphanumeric identifier **600** conspicuously preprinted on a sidewall panel of the container **100**, and the radio frequency tags **102** have a unique radio frequency identifier **604** (the identifier is visible on the drawings to illustrate the existence of thereof) stored thereon. The optical identifier **600** may be associated with a waste container user information **608**, such as a name, address, user identifier number, account number, etc., or may require such an association during deployment.

A handheld computer **612** for use in the method has an optical character recognition reader **616** for reading the optical identifier **600** from the waste container **100** and a communications module for receiving the radio frequency identifier **604** from the radio frequency tag **102**. In FIG. **21**, the handheld computer **612** has converted the optical identifier **600** and the radio frequency identifier **604** to an electronic format and associated both with user information **608**. The user information **608** can be stored at the external site **300**, on the server **302** or other computer **304**, either of which can electronically access the database **306** having user information stored therein. The user information **608** can also be stored on the memory on the handheld computer **612**.

FIG. **22** shows the handheld computer **612** in communication with the external site **300** discussed above. Software routines or modules are executable to create entries in a database **306** corresponding to an association between the optical identifiers **600**, the radio frequency identifiers **604**, and the user information data **608**. The paired or matched information is stored on the database **306** for future use consistent with the principles discussed above. The software modules may be store in the memory of the handheld computer **612** and executable therefrom or stored at the external site **300** and downloadable or executable therefrom.

According to invention, the optical character recognition reader **616** is aimed at the waste container **100** which has the unique optical waste container identifier **600** printed thereon. The unique waste container optical identifier **600** is read using the optical character recognition reader **616**. An executable software routine or module stored in the handheld computer converts the unique waste container optical identifier **600** to a unique electronically formatted waste container identifier. The unique radio frequency signal identifier **604** stored on the radio frequency emitting tag **102** is associated with the unique electronically formatted waste container identifier. The radio frequency emitting tag **102** is attached to the waste container **102**.

For example, the unique electronically formatted waste container identifier is stored on the optical character recognition reader **616**.

For example, the unique radio frequency signal identifier **604** is read from the radio frequency emitting tag **102**.

For example, a software module for associating the unique radio frequency signal identifier **604** with the unique electronically formatted waste container identifier stored on the optical character recognition reader **616** is activated. The

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software module may be stored on the memory of the handheld computer **612** or accessible from the external site **300**.

For example, a software module for associating the unique radio frequency signal identifier **604** with personal user data **608** of a user of the waste container **100** is activated. The software module may be stored on the memory of the handheld computer **612** or accessible from the external site **300**.

For example, a software module for generating a unique database entry comprising the unique radio frequency signal identifier **604** and the personal user data **608** associated therewith may be activated. The software module may be stored on the memory of the handheld computer **612** or accessible from the external site **300**.

For example, the optical character recognition reader **616** and the radio frequency signal identifier reader for reading the unique radio frequency signal identifier **604** from the radio frequency emitting tag **102** are included on the handheld computer **612**.

For example, the association of the unique radio frequency signal identifier **604** with the unique electronically formatted waste container identifier is performed by a software module executed on the handheld computer **612**.

For example, the handheld computer **s** used to transmit the unique radio frequency signal identifier **604** and the unique electronically formatted waste container identifier associated therewith to an external site **300**, preferably a central server **302** and/or computer **304**.

For example, the waste container **100** having the unique optical waste container identifier **600** pre-printed thereon is distributed to the user of the waste container.

In another embodiment, the optical character recognition reader **616** capable of reading the unique optical waste container identifier **600** printed on the waste container is provided. The reader **616** is capable of converting the unique optical waste container identifier **600** to a unique electronically formatted waste container identifier. The radio frequency emitting tag **102** with the unique radio frequency signal identifier **604** is also provided. A software routine or module for associates the unique electronically formatted waste container identifier with the radio frequency emitting tag identifier **604**. A further software routine or module associates the radio frequency signal identifier **604** with personal information data **608** associated with a user of the waste container **100**. A further software routine or module compiles a database comprising the personal information data **608** associated with the user of the waste container **100**, the unique radio frequency signal identifier **604**, and unique electronically formatted waste container identifier.

For example, the radio frequency emitting tag reader reads the radio frequency signal identifier **604** on the radio frequency emitting tag **102**.

For example, the handheld computer **612** includes the radio frequency emitting tag reader and the optical character recognition reader **616**.

For example, the software routine or module for associating the unique electronically formatted waste container identifier with the radio frequency emitting tag identifier **604**

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is stored in the memory located on the handheld computer **612** and is executable thereon.

For example, the software routine or module for associating the radio frequency signal identifier **604** with personal information data **608** associated with the user of the waste container **100** is stored in the memory located on the handheld computer **612**.

While the specific embodiments have been illustrated and described, numerous modifications come to mind without significantly departing from the spirit of the invention, and the scope of protection is only limited by the scope of the accompanying Claims.

What is claimed is:

1. A waste removal method for use with an external receptacle comprising a radio frequency identification tag containing information about a customer assigned to the receptacle, the method comprising:

providing a waste removal vehicle comprising

a waste bin having a chamber and an opening to the chamber for receiving and transporting waste material from the external receptacle,

at least one receiver,

a controller on the waste removal vehicle in electronic communication with the at least one receiver, and

a mechanism on the waste removal vehicle in electronic communication with the controller, for transferring the external receptacle along a path from a first position to a second position;

providing to the controller information from the radio frequency identification tag from at least one of the at least one receivers;

sending the information from the radio frequency identification tag from the controller to an external database containing information about the customer;

receiving, via the controller, a signal from the external database of a pass condition or a fail condition;

in response to receiving the signal of the pass condition, activating the mechanism via the controller, or

in response to receiving the signal of the fail condition, deactivating the mechanism via the controller.

2. The method of claim 1, wherein the second position is proximal the opening.

3. The method of claim 1, wherein the pass condition signal is generated in response to determining with a processor in the external database that the customer has an account in good standing.

4. The method of claim 1, further comprising activating the first receiver.

5. The method of claim 4, wherein the activation is performed in response to a signal from a proximity switch, a photo sensor or a mechanical switch.

6. The method of claim 1, further comprising activating the first receiver upon contact by the mechanism for transferring with the external receptacle.

7. The method of claim 1, wherein the at least one receiver comprises a first receiver located at first position and a second receiver located at a second position spaced from the first position.

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